Profile Documentation

Profile namespace: `http://opensource.ieee.org/emtiop01p01#`

Concrete Classes

ACDCConverterDCTerminal

DC

A DC electrical connection point at the AC/DC converter. The AC/DC converter is electrically connected also to the AC side. The AC connection is inherited from the AC conducting equipment in the same way as any other AC equipment. The AC/DC converter DC terminal is separate from generic DC terminal to restrict the connection with the AC side to AC/DC converter and so that no other DC conducting equipment can be connected to the AC side.

Native Members

polarity	0..1	DCPolarityKind	Represents the normal network polarity condition. Depending on the converter configuration the value shall be set as follows: - For a monopole with two converter terminals use DCPolarityKind "positive" and "negative".

			- For a bi-pole or symmetric monopole with three converter terminals use DCPolarityKind "positive", "middle" and "negative".
DCConductingEquipment	0..1	ACDCConverter	A DC converter terminal belong to an DC converter.

Inherited Members

DCNode	1..1	DCNode	see DCBaseTerminal
sequenceNumber	1..1	Integer	see ACDCTerminal
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

ACLineSegment

Wires

A line segment is a conductor or combination of conductors, with consistent electrical characteristics along its length, building a single electrical system that carries alternating current between two points in the power system.

The BaseVoltage at the two ends of a line segment shall have the same BaseVoltage.nominalVoltage. However, boundary lines may have slightly different BaseVoltage.nominalVoltages and variation is allowed. Larger voltage difference in general requires use of an equivalent branch.

Line segment impedances can be either directly described in electrical terms or physical line detail can be provided from which impedances can be calculated.

Directly described impedances

For symmetrical, transposed three phase line segments, it is sufficient to use attributes of the line segment, which describe impedances and admittances for the entire length of the line segment. Additionally, line segment impedances can be computed by using line segment length and associated per length impedances.

Unbalanced modeling of impedances is supported by the per length phase impedance matrix (PerLengthPhaseImpedance) in conjunction with phase-to-sequence number mapping supplied by either ACLineSegmentPhase or WirePosition. The sequence numbers are referenced by the row and column attributes of the per length phase impedance matrix. This method enables single-phase and two-phase line segments, and transpositions of phases, to be described using the same per length phase impedance matrix. The length of the line segment is used in the computation of total impedance values for the line segment.

Line detail characteristics

There are three approaches to providing line detail and all use WireAssembly to supply line positions:

- Option 1 - WireAssembly supplies only line positions. ACLineSegmentPhase points to wire type and intraphase spacing and supplies the phase-to-sequence number mapping.
- Option 2 - WireAssembly supplies line position and, for each position, also supplies wire type and intraphase spacing. ACLineSegmentPhase supplies the phase-to-sequence number mapping.

- Option 3 - WireAssembly supplies line position and, for each position, also supplies wire type and intraphase spacing and phase. WireAssembly therefore supplies the phase-to-sequence number mapping and ACLineSegmentPhase is not needed.

Native Members

b0ch	1..1	Susceptance	Zero sequence shunt (charging) susceptance, uniformly distributed, of the entire line segment.
bch	1..1	Susceptance	Positive sequence shunt (charging) susceptance, uniformly distributed, of the entire line segment. This value represents the full charging over the full length of the line segment.
r	1..1	Resistance	Positive sequence series resistance of the entire line segment.
r0	1..1	Resistance	Zero sequence series resistance of the entire line segment.
x	1..1	Reactance	Positive sequence series reactance of the entire line segment.
x0	1..1	Reactance	Zero sequence series reactance of the entire line segment.

Inherited Members

length	1..1	Length	see Conductor
BaseVoltage	0..1	BaseVoltage	see ConductingEquipment
inService	1..1	Boolean	see Equipment
EquipmentContainer	1..1	EquipmentContainer	see Equipment
mRID	1..1	String	see PowerSystemResource
name	1..1	String	see PowerSystemResource
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

ACPointOfCommonCoupling Core

Point of interconnection of the DC converter station to the adjacent AC system (IEC 60633).

Native Members

ConnectivityNode	0..1	ConnectivityNode	Connectivity node which is a point of common coupling AC.

Inherited Members

mRID	1..1	String	see PointOfCommonCoupling
name	1..1	String	see PointOfCommonCoupling
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

ApparentPowerLimit

OperationalLimits

Apparent power limit.

Native Members

value	0..1	ApparentPower	The apparent power limit. The attribute shall be a positive value or zero.

Inherited Members

mRID	1..1	String	see OperationalLimit
name	1..1	String	see OperationalLimit
OperationalLimitSet	0..1	OperationalLimitSet	see OperationalLimit
OperationalLimitType	0..1	OperationalLimitType	see OperationalLimit
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

AsynchronousMachine

Wires

A rotating machine whose shaft rotates asynchronously with the electrical field. Also known as an induction machine with no external connection to the rotor windings, e.g. squirrel-cage induction machine.

Native Members

asynchronousMachineType	0..1	AsynchronousMachineKind	Indicates the type of Asynchronous Machine (motor or generator).
converterFedDrive	0..1	Boolean	Indicates whether the machine is a converter fed drive. Used for short circuit data exchange according to IEC 60909.
efficiency	0..1	PerCent	Efficiency of the asynchronous machine at nominal operation as a percentage. Indicator for converter drive motors. Used for short circuit data exchange according to IEC 60909.
iaIrRatio	0..1	Float	Ratio of locked-rotor current to the rated current of the motor (Ia/Ir). Used for short circuit data exchange according to IEC 60909.
nominalFrequency	0..1	Frequency	Nameplate data indicates if the machine is 50 Hz or 60 Hz.

nominalSpeed	0..1	RotationSpeed	Nameplate data. Depends on the slip and number of pole pairs.
polePairNumber	0..1	Integer	Number of pole pairs of stator. Used for short circuit data exchange according to IEC 60909.
ratedMechanicalPower	0..1	ActivePower	Rated mechanical power (Pr in IEC 60909-0). Used for short circuit data exchange according to IEC 60909.
reversible	0..1	Boolean	Indicates for converter drive motors if the power can be reversible. Used for short circuit data exchange according to IEC 60909.
rr1	0..1	Resistance	Damper 1 winding resistance.
rr2	0..1	Resistance	Damper 2 winding resistance.
rxLockedRotorRatio	0..1	Float	Locked rotor ratio (R/X). Used for short circuit data exchange according to IEC 60909.
tpo	0..1	Seconds	

			Transient rotor time constant (greater than tppo).
tppo	0..1	Seconds	Sub-transient rotor time constant (greater than 0).
xlr1	0..1	Reactance	Damper 1 winding leakage reactance.
xlr2	0..1	Reactance	Damper 2 winding leakage reactance.
xm	0..1	Reactance	Magnetizing reactance.
xp	0..1	Reactance	Transient reactance (unsaturated) (greater than or equal to xpp).
xpp	0..1	Reactance	Sub-transient reactance (unsaturated).
xs	0..1	Reactance	Synchronous reactance (greater than xp).

Inherited Members

p	1..1	ActivePower	see RotatingMachine
q	1..1	ReactivePower	see RotatingMachine
ratedS	1..1	ApparentPower	see RotatingMachine
ratedU	1..1	Voltage	see RotatingMachine
GeneratingUnit	1..1	GeneratingUnit	see RotatingMachine
BaseVoltage	0..1	BaseVoltage	see ConductingEquipment
inService	1..1	Boolean	see Equipment
EquipmentContainer	1..1	EquipmentContainer	see Equipment
mRID	1..1	String	see PowerSystemResource
name	1..1	String	see PowerSystemResource

mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

AsynchronousMachineTimeConstantReactance

AsynchronousMachineDynamics

Parameter details:

1. If *X''* = *X'*, a single cage (one equivalent rotor winding per axis) is modelled.
2. The â€œ*p*â€� in the attribute names is a substitution for a â€œprimeâ€� in the usual parameter notation, e.g. *tpo* refers to *T'o*.

The parameters used for models expressed in time constant reactance form include:

- RotatingMachine.ratedS (*MVAbase*);

- RotatingMachineDynamics.damping (*D*);

- RotatingMachineDynamics.inertia (*H*);

- RotatingMachineDynamics.saturationFactor (*S1*);

- RotatingMachineDynamics.saturationFactor120 (*S12*);

- RotatingMachineDynamics.statorLeakageReactance (*Xl*);

- RotatingMachineDynamics.statorResistance (*Rs*);

- .xs (*Xs*);

- .xp (*X′*);

- .xpp (*X″*);

- .tpo (*T′o*);

- .tppo (*T″o*).

Native Members

tpo	0..1	Seconds	Transient rotor time constant (*T′o*) (> AsynchronousMachineTimeConstantReactance.tppo). Typical value = 5.
tppo	0..1	Seconds	Subtransient rotor time constant (*T″o*) (> 0). Typical value = 0,03.
xp	0..1	PU	Transient reactance (unsaturated) (*X′*) (>= AsynchronousMachineTimeConstantReactance.xpp). Typical value = 0,5.
xpp	0..1	PU	Subtransient reactance (unsaturated) (*X″*) (> RotatingMachineDynamics.statorLeakageReactance). Typical value = 0,2.
xs	0..1	PU	

			Synchronous reactance (*Xs*) (>= AsynchronousMachineTimeConstantReactance.xp). Typical value = 1,8.

Inherited Members

AsynchronousMachine	0..1	AsynchronousMachine	see AsynchronousMachineDynamics
damping	1..1	Float	see RotatingMachineDynamics
inertia	1..1	Seconds	see RotatingMachineDynamics
statorLeakageReactance	1..1	PU	see RotatingMachineDynamics
statorResistance	1..1	PU	see RotatingMachineDynamics
mRID	1..1	String	see DynamicsFunctionBlock
enabled	1..1	Boolean	see DynamicsFunctionBlock
name	1..1	String	see DynamicsFunctionBlock

mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

BaseVoltage

Core

Defines a system base voltage which is referenced. This may be different than the rated voltage.

Native Members

mRID	1..1	String	Master resource identifier issued by a model authority. The mRID is unique within an exchange context. Global uniqueness is easily achieved by using a UUID, as specified in IETF RFC 4122, for the mRID. The use of UUID is strongly recommended. For CIMXML data files in RDF syntax conforming to IEC 61970-552, the mRID is mapped to rdf:ID or rdf:about attributes that identify CIM object elements.

name	1..1	String	The name is any free human readable and possibly non unique text naming the object.
nominalVoltage	1..1	Voltage	The power system resource's base voltage, expressed on a phase-to-phase (line-to-line) basis. Shall be a positive value and not zero.

Inherited Members

mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

BatteryUnit

Production

An electrochemical energy storage device.

Native Members

batteryState	1..1	BatteryStateKind	

			The current state of the battery (charging, full, etc.).
ratedE	1..1	RealEnergy	Full energy storage capacity of the battery. The attribute shall be a positive value.
storedE	1..1	RealEnergy	Amount of energy currently stored. The attribute shall be a positive value or zero and lower than BatteryUnit.ratedE.

Inherited Members

maxP	1..1	ActivePower	see PowerElectronicsUnit
minP	1..1	ActivePower	see PowerElectronicsUnit
PowerElectronicsConnection	1..1	PowerElectronicsConnection	see PowerElectronicsUnit
inService	1..1	Boolean	see Equipment
EquipmentContainer	1..1	EquipmentContainer	see Equipment

mRID	1..1	String	see PowerSystemResource
name	1..1	String	see PowerSystemResource
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

ConnectivityNode

Core

Connectivity nodes are points where terminals of AC conducting equipment are connected together with zero impedance.

Native Members

mRID	1..1	String	Master resource identifier issued by a model authority. The mRID is unique within an exchange context. Global uniqueness is easily achieved by using a UUID, as specified in IETF RFC 4122, for the mRID. The use of UUID is strongly recommended.

			For CIMXML data files in RDF syntax conforming to IEC 61970-552, the mRID is mapped to rdf:ID or rdf:about attributes that identify CIM object elements.
name	1..1	String	The name is any free human readable and possibly non unique text naming the object.
ConnectivityNodeContainer	1..1	ConnectivityNodeContainer	Container of this connectivity node.
TopologicalNode	0..1	TopologicalNode	The topological node to which this connectivity node is assigned. May depend on the current state of switches in the network.

Inherited Members

mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

CsConverter

DC side of the current source converter (CSC).

The firing angle controls the dc voltage at the converter, both for rectifier and inverter. The difference between the dc voltages of the rectifier and inverter determines the dc current. The extinction angle is used to limit the dc voltage at the inverter, if needed, and is not used in active power control. The firing angle, transformer tap position and number of connected filters are the primary means to control a current source dc line. Higher level controls are built on top, e.g. DC voltage, dc current and active power. From a steady state perspective it is sufficient to specify the desired active power transfer (ACDCConverter.targetPpcc) and the control functions will set the dc voltage, dc current, firing angle, transformer tap position and number of connected filters to meet this. Therefore attributes targetAlpha and targetGamma are not applicable in this case.

Attributes targetAlpha and targetGamma are mutually exclusive therefore only one of them can be defined to describe an operating target.

The reactive power consumed by the converter is a function of the firing angle, transformer tap position and number of connected filter, which can be approximated with half of the active power. The losses are a function of the dc voltage and dc current.

The attributes minAlpha and maxAlpha define the range of firing angles for rectifier operation between which no discrete tap changer action takes place. The range is typically 10 to 18 degrees.

The attributes minGamma and maxGamma define the range of extinction angles for inverter operation between which no discrete tap changer action takes place. The range is typically 17 to 20 degrees.

Native Members

alpha	0..1	AngleDegrees	Firing angle that determines the DC voltage at the converter DC terminal. Typical value between 10 degrees and 18 degrees for a rectifier. It is converter's state variable,

			result from power flow. The attribute shall be a positive value.
gamma	0..1	AngleDegrees	Extinction angle. It is used to limit the DC voltage at the inverter if needed. Typical value between 17 degrees and 20 degrees for an inverter. It is converter's state variable, result from power flow. The attribute shall be a positive value.
maxAlpha	0..1	AngleDegrees	Maximum firing angle. It is the converter's configuration data used in power flow. The attribute shall be a positive value.
maxGamma	0..1	AngleDegrees	Maximum extinction angle. It is the converter's configuration data used in power flow. The attribute shall be a positive value.
maxIdc	0..1	CurrentFlow	The maximum direct current (Id) on the DC side at which the converter should operate. It is the converter's configuration data use in power flow. The attribute shall be a positive value.
minAlpha	0..1	AngleDegrees	Minimum firing angle. It is the converter's configuration data used in power flow. The attribute shall be a positive value.

minGamma	0..1	AngleDegrees	Minimum extinction angle. It is the converter's configuration data used in power flow. The attribute shall be a positive value.
minIdc	0..1	CurrentFlow	The minimum direct current (Id) on the DC side at which the converter should operate. It is the converter's configuration data used in power flow. The attribute shall be a positive value.
operatingMode	0..1	CsOperatingModeKind	Indicates whether the DC pole is operating as an inverter or as a rectifier. It is converter's control variable used in power flow.
pPccControl	0..1	CsPpccControlKind	Kind of active power control.
ratedIdc	0..1	CurrentFlow	Rated converter DC current, also called IdN. The attribute shall be a positive value. It is the converter's configuration data used in power flow.
targetAlpha	0..1	AngleDegrees	Target firing angle. It is converter's control variable used in power flow. It is only applicable for rectifier control. Allowed values are within the range

			minAlpha<=targetAlpha<=maxAlpha. The attribute shall be a positive value.
targetGamma	0..1	AngleDegrees	Target extinction angle. It is converter's control variable used in power flow. It is only applicable for inverter control. Allowed values are within the range minGamma<=targetGamma<=maxGamma. The attribute shall be a positive value.
targetIdc	0..1	CurrentFlow	DC current target value. It is converter's control variable used in power flow. The attribute shall be a positive value.

Inherited Members

baseS	0..1	ApparentPower	see ACDCConverter
idc	0..1	CurrentFlow	see ACDCConverter
idleLoss	0..1	ActivePower	see ACDCConverter
maxP	0..1	ActivePower	see ACDCConverter

maxUdc	0..1	Voltage	see ACDCConverter
minP	0..1	ActivePower	see ACDCConverter
minUdc	0..1	Voltage	see ACDCConverter
numberOfValves	0..1	Integer	see ACDCConverter
p	0..1	ActivePower	see ACDCConverter
poleLossP	0..1	ActivePower	see ACDCConverter
q	0..1	ReactivePower	see ACDCConverter
ratedUdc	0..1	Voltage	see ACDCConverter
resistiveLoss	0..1	Resistance	see ACDCConverter
switchingLoss	0..1	ActivePowerPerCurrentFlow	see ACDCConverter
targetPpcc	0..1	ActivePower	see ACDCConverter

targetUdc	0..1	Voltage	see ACDCConverter
uc	0..1	Voltage	see ACDCConverter
udc	0..1	Voltage	see ACDCConverter
valveU0	0..1	Voltage	see ACDCConverter
BaseVoltage	0..1	BaseVoltage	see ConductingEquipment
inService	1..1	Boolean	see Equipment
EquipmentContainer	1..1	EquipmentContainer	see Equipment
mRID	1..1	String	see PowerSystemResource
name	1..1	String	see PowerSystemResource
mRID	0..1	String	see IdentifiedObject
name	0..1	String	

see IdentifiedObject

CurveData

Core

Multi-purpose data points for defining a curve. The use of this generic class is discouraged if a more specific class can be used to specify the X and Y axis values along with their specific data types.

Native Members

xvalue	1..1	Float	The data value of the X-axis variable, depending on the X-axis units.
y1value	1..1	Float	The data value of the first Y-axis variable, depending on the Y-axis units.
Curve	1..1	Curve	The curve of this curve data point.

DCBreaker

DC

A breaker within a DC system.

Inherited Members

locked	0..1	Boolean	see DCSwitch
normalOpen	0..1	Boolean	see DCSwitch
open	0..1	Boolean	see DCSwitch
retained	0..1	Boolean	see DCSwitch
ratedCurrent	0..1	CurrentFlow	see DCConductingEquipment
ratedUdc	0..1	Voltage	see DCConductingEquipment
inService	1..1	Boolean	see Equipment
EquipmentContainer	1..1	EquipmentContainer	see Equipment
mRID	1..1	String	see PowerSystemResource

name	1..1	String	see PowerSystemResource
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

DCDisconnector

DC

A disconnector within a DC system.

Inherited Members

locked	0..1	Boolean	see DCSwitch
normalOpen	0..1	Boolean	see DCSwitch
open	0..1	Boolean	see DCSwitch
retained	0..1	Boolean	see DCSwitch

ratedCurrent	0..1	CurrentFlow	see DCConductingEquipment
ratedUdc	0..1	Voltage	see DCConductingEquipment
inService	1..1	Boolean	see Equipment
EquipmentContainer	1..1	EquipmentContainer	see Equipment
mRID	1..1	String	see PowerSystemResource
name	1..1	String	see PowerSystemResource
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

DCEnergySource

Emtiop

A source of DC power that is independent of the AC system, e.g., a battery or solar panel. The internal representation may include current sources, voltage sources, diodes, etc. Use DCSourceKind to provide guidance on the internal representation.

Native Members

kind	0..1	DCSourceKind	
p	0..1	ActivePower	The power output, negative for load or charging.
pMax	0..1	ActivePower	Maximum power available from the primary source, e.g., photovoltaic panels or a battery.
pMaxLoad	0..1	ActivePower	Maximum load or battery charging power.
pMin	0..1	ActivePower	Minimum power available from the supply.
pMinLoad	0..1	ActivePower	Minimum load or battery charging power.
rSeries	0..1	Resistance	Series source resistance.

rShunt	0..1	Resistance	Shunt source resistance.

Inherited Members

ratedCurrent	0..1	CurrentFlow	see DCConductingEquipment
ratedUdc	0..1	Voltage	see DCConductingEquipment
inService	1..1	Boolean	see Equipment
EquipmentContainer	1..1	EquipmentContainer	see Equipment
mRID	1..1	String	see PowerSystemResource
name	1..1	String	see PowerSystemResource
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

DCEquipmentContainer

DC

A modelling construct to provide a root class for containment of DC as well as AC equipment. The class differ from the EquipmentContainer for AC in that it may also contain DCNode(-s). Hence it can contain both AC and DC equipment.

Inherited Members

mRID	1..1	String	see PowerSystemResource
name	1..1	String	see PowerSystemResource
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

DCGround

DC

A ground within a DC system.

Native Members

inductance	0..1	Inductance	Inductance to ground.
r	0..1	Resistance	Resistance to ground.

Inherited Members

ratedCurrent	0..1	CurrentFlow	see DCConductingEquipment
ratedUdc	0..1	Voltage	see DCConductingEquipment
inService	1..1	Boolean	see Equipment
EquipmentContainer	1..1	EquipmentContainer	see Equipment
mRID	1..1	String	see PowerSystemResource
name	1..1	String	see PowerSystemResource
mRID	0..1	String	see IdentifiedObject

name	0..1	String	see IdentifiedObject

DCLineSegment

DC

A wire or combination of wires not insulated from one another, with consistent electrical characteristics, used to carry direct current between points in the DC region of the power system.

Native Members

capacitance	0..1	Capacitance	Capacitance of the DC line segment. Significant for cables only.
inductance	0..1	Inductance	Inductance of the DC line segment. Negligible compared with DCSeriesDevice used for smoothing.
length	0..1	Length	Segment length for calculating line section capabilities.
resistance	0..1	Resistance	Resistance of the DC line segment.

Inherited Members

ratedCurrent	0..1	CurrentFlow	see DCConductingEquipment
ratedUdc	0..1	Voltage	see DCConductingEquipment
inService	1..1	Boolean	see Equipment
EquipmentContainer	1..1	EquipmentContainer	see Equipment
mRID	1..1	String	see PowerSystemResource
name	1..1	String	see PowerSystemResource
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

DCNode

DC

DC nodes are points where terminals of DC conducting equipment are connected together with zero impedance.

Native Members

mRID	1..1	String	Master resource identifier issued by a model authority. The mRID is unique within an exchange context. Global uniqueness is easily achieved by using a UUID, as specified in IETF RFC 4122, for the mRID. The use of UUID is strongly recommended. For CIMXML data files in RDF syntax conforming to IEC 61970-552, the mRID is mapped to rdf:ID or rdf:about attributes that identify CIM object elements.
name	1..1	String	The name is any free human readable and possibly non unique text naming the object.
DCEquipmentContainer	0..1	DCEquipmentContainer	The DC container for the DC nodes.
DCTopologicalNode	0..1	DCTopologicalNode	The DC topological node to which this DC connectivity node is assigned. May depend on the

current state of switches in the network.

Inherited Members

mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

DCSeriesDevice DC

A series device within the DC system, typically a reactor used for filtering or smoothing. Needed for transient and short circuit studies.

Native Members

inductance	0..1	Inductance	Inductance of the device.
resistance	0..1	Resistance	Resistance of the DC device.

Inherited Members

ratedCurrent	0..1	CurrentFlow	see DCConductingEquipment
ratedUdc	0..1	Voltage	see DCConductingEquipment
inService	1..1	Boolean	see Equipment
EquipmentContainer	1..1	EquipmentContainer	see Equipment
mRID	1..1	String	see PowerSystemResource
name	1..1	String	see PowerSystemResource
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

DCShunt

DC

A shunt device within the DC system, typically used for filtering. Needed for transient and short circuit studies.

Native Members

capacitance	0..1	Capacitance	Capacitance of the DC shunt.
resistance	0..1	Resistance	Resistance of the DC device.

Inherited Members

ratedCurrent	0..1	CurrentFlow	see DCConductingEquipment
ratedUdc	0..1	Voltage	see DCConductingEquipment
inService	1..1	Boolean	see Equipment
EquipmentContainer	1..1	EquipmentContainer	see Equipment
mRID	1..1	String	see PowerSystemResource
name	1..1	String	see PowerSystemResource

mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

DCTerminal

DC

An electrical connection point to generic DC conducting equipment.

Native Members

polarity	0..1	DCTerminalPolarityKind	Represents the normal network polarity condition. Used in DC system configurations that have explicit polarity of the terminals, e.g., voltage source converter (VSC) technology.
DCConductingEquipment	0..1	DCConductingEquipment	An DC terminal belong to a DC conducting equipment.

Inherited Members

DCNode	1..1	DCNode	see DCBaseTerminal
sequenceNumber	1..1	Integer	see ACDCTerminal
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

DCTopologicalNode

DC

DC bus.

Inherited Members

mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

DetailedModelDynamics

The main class that packages all related to this detailed model. This includes all parameters, functions, signals, etc.

Native Members

DetailedModelTypeDynamics	0..1	DetailedModelTypeDynamics	The type of detailed model dynamics that is applied to the detailed model dynamics.
DynamicsFunctionBlock	0..1	DynamicsFunctionBlock	The dynamics function block for this detailed model dynamics.
Equipment	0..1	Equipment	The equipment which behaviour this detailed model dynamics represents.

Inherited Members

mRID	1..1	String	see DynamicsFunctionBlock
enabled	1..1	Boolean	see DynamicsFunctionBlock

name	1..1	String	see DynamicsFunctionBlock
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

DiagramObjectPoint

DiagramLayout

A point in a given space defined by 3 coordinates and associated to a diagram object. The coordinates may be positive or negative as the origin does not have to be in the corner of a diagram.

Native Members

sequenceNumber	1..1	Integer	The sequence position of the point, used for defining the order of points for diagram objects acting as a polyline or polygon with more than one point. The attribute shall be a positive value.
xPosition	1..1	Float	The X coordinate of this point.

yPosition	1..1	Float	The Y coordinate of this point.
DiagramObject	1..1	DiagramObject	The diagram object with which the points are associated.

DisconnectingCircuitBreaker

Wires

A circuit breaking device including disconnecting function, eliminating the need for separate disconnectors.

Inherited Members

BaseVoltage	0..1	BaseVoltage	see ConductingEquipment
inService	1..1	Boolean	see Equipment
EquipmentContainer	1..1	EquipmentContainer	see Equipment
mRID	1..1	String	see PowerSystemResource
name	1..1	String	see PowerSystemResource

mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

EnergyConsumer

Wires

Generic user of energy - a point of consumption on the power system model.

EnergyConsumer.pfixed, .qfixed, .pfixedPct and .qfixedPct have meaning only if there is no LoadResponseCharacteristic associated with EnergyConsumer or if LoadResponseCharacteristic.exponentModel is set to False.

Native Members

p	1..1	ActivePower	Active power of the load. Load sign convention is used, i.e. positive sign means flow out from a node. For voltage dependent loads the value is at rated voltage. Starting value for a steady state solution.

q	1..1	ReactivePower	Reactive power of the load. Load sign convention is used, i.e. positive sign means flow out from a node. For voltage dependent loads the value is at rated voltage. Starting value for a steady state solution.
LoadResponse	1..1	LoadResponseCharacteristic	The load response characteristic of this load. If missing, this load is assumed to be constant power.

Inherited Members

BaseVoltage	0..1	BaseVoltage	see ConductingEquipment
inService	1..1	Boolean	see Equipment
EquipmentContainer	1..1	EquipmentContainer	see Equipment
mRID	1..1	String	see PowerSystemResource

name	1..1	String	see PowerSystemResource
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

EnergySource

Wires

A generic equivalent for an energy supplier on a transmission or distribution voltage level.

Native Members

nominalVoltage	1..1	Voltage	Phase-to-phase nominal voltage.
r	1..1	Resistance	Positive sequence Thevenin resistance.
r0	1..1	Resistance	Zero sequence Thevenin resistance.
voltageAngle	1..1	AngleRadians	Phase angle of a-phase open circuit used when voltage characteristics need to be imposed at the node

			associated with the terminal of the energy source, such as when voltages and angles from the transmission level are used as input to the distribution network. The attribute shall be a positive value or zero.
voltageMagnitude	1..1	Voltage	Phase-to-phase open circuit voltage magnitude used when voltage characteristics need to be imposed at the node associated with the terminal of the energy source, such as when voltages and angles from the transmission level are used as input to the distribution network. The attribute shall be a positive value or zero.
x	1..1	Reactance	Positive sequence Thevenin reactance.
x0	1..1	Reactance	Zero sequence Thevenin reactance.

Inherited Members

BaseVoltage	0..1	BaseVoltage	see ConductingEquipment

inService	1..1	Boolean	see Equipment
EquipmentContainer	1..1	EquipmentContainer	see Equipment
mRID	1..1	String	see PowerSystemResource
name	1..1	String	see PowerSystemResource
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

EquipmentContainer

Core

A modelling construct to provide a root class for containing equipment.

Inherited Members

mRID	1..1	String	see PowerSystemResource

name	1..1	String	see PowerSystemResource
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

HydroGeneratingUnit

Production

A generating unit whose prime mover is a hydraulic turbine (e.g. Francis, Pelton, Kaplan).

Inherited Members

maxOperatingP	1..1	ActivePower	see GeneratingUnit
minOperatingP	1..1	ActivePower	see GeneratingUnit
inService	1..1	Boolean	see Equipment
EquipmentContainer	1..1	EquipmentContainer	see Equipment

mRID	1..1	String	see PowerSystemResource
name	1..1	String	see PowerSystemResource
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

IBRPlant

Emtiop

An inverter-based resource (IBR) plant comprising a collection of components, e.g., a PowerElectronicsConnection with associated controls and GeneratingUnit, one or more PowerTransformers, one or more DisconnectingCircuitBreakers, and one or more ACLineSegments. The components may also include AC filter and DC bus modeling.

Native Members

dcLinkVoltage	0..1	Voltage	Voltage of the DC bus, used to scale average source models or to supply switching models of the converter.
switchingFrequency	0..1	Frequency	Pulse width modulation (PWM) switching frequency of the firing

pulses in a switching model of the converter.

Inherited Members

ACPointOfCommonCoupling	0..1	ACPointOfCommonCoupling	see ConnectedFacility
Equipments	0..unbounded	Equipment	see ConnectedFacility
mRID	1..1	String	see PowerSystemResource
name	1..1	String	see PowerSystemResource
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

IEEECigreAPI

Emtiop

A dynamic link library (DLL) or other application programming interface (API) for inverter-based resource (IBR) control and other control applications, as defined in CIGRE Technical Brochure TB 958 and IEEE Standards

Association P3597. Attributes prefixed by **api** correspond to members of the IEEE_Cigre_DLLInterface_Model_Info header file structure that was documented in TB 958; they should be obtained and verified using the API.

Native Members

apiDLLInterfaceVersion	0..1	String	Codifies a four-number version of the CIGRE TB 958 interface standard supported by this model, in string format, as returned from the API.
apiEmtRmsMode	0..1	IEEECigreAPIModeKind	Identify whether the model runs in EMT, RMS, or both kinds of simulation, as returned from the CIGRE TB 958 API.
apiFixedStepBaseSampleTime	0..1	Seconds	Hard-coded simulation time step for this model, as returned from the CIGRE TB 958 API.
apiModelName	0..1	String	The ModelName as returned from the CIGRE TB 958 API. Not necessarily equal to the inherited IdentifiedObject.name.
apiModelVersion	0..1	String	Version of this model instance, as returned from the CIGRE TB 958 API.

shareable	0..1	Boolean	True if this DLL can be loaded and used by different instances of Equipment in the same simulation. This information is not available from the DLL API; it must be determined from careful review of the DLL documentation.
snapshotUri	0..1	String	Location of the optional snapshot file for initializing the DLL from a saved state. Either a universal resource identifier or network-accessible filename. It is not obtainable from the DLL API.
uri	0..1	String	Location of the DLL, e.g., a universal resource identifier or network-accessible filename. It is not obtainable from the DLL API.
IEEECigreAPIInfo	0..1	IEEECigreAPIInfo	Expanded set of attributes available from the CIGRE TB 958 API.

Inherited Members

DetailedModelTypeDynamics	0..1	DetailedModelTypeDynamics	see DetailedModelDynamics

DynamicsFunctionBlock	0..1	DynamicsFunctionBlock	see DetailedModelDynamics
Equipment	0..1	Equipment	see DetailedModelDynamics
mRID	1..1	String	see DynamicsFunctionBlock
enabled	1..1	Boolean	see DynamicsFunctionBlock
name	1..1	String	see DynamicsFunctionBlock
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

IEEECigreAPIInfo

Emtiop

Supplemental information about this interface from the CIGRE TB 958 API.

Native Members

apiGeneralInformation	0..1	String	One of two general description fields; also see the apiModelDescription attribute.
apiModelCreated	0..1	DateTime	Date and time of the first model (not API) version.
apiModelCreator	0..1	String	Identifies the person or the organization who first created the model.
apiModelDescription	0..1	String	One of two general description fields; also see the apiGeneralDescription attribute.
apiModelLastModifiedBy	0..1	String	Identifies the person or the organization who first created the model.
apiModelLastModifiedDate	0..1	DateTime	Date and time of the latest model version.
apiModelModifiedComment	0..1	String	A description of the latest model update.

apiModelModifiedHistory	0..1	String	A history of model updates; may be a change log or other multi-paragraph text.
apiNumDoubleStates	0..1	Integer	Size of internal double-precision array storage needed by the model instance for state variables. These are not represented in CIM, but the information could help identify different versions of this model. The EMT/RMS simulator manages this memory.
apiNumFloatStates	0..1	Integer	Size of internal single-precision array storage needed by this model for state variables. These are not represented in CIM, but the information could help identify different versions of this model. The EMT/RMS simulator manages this memory.
apiNumInputPorts	0..1	Integer	The number of input ports expected by this model, which should match cardinality of the associated IEEECigreAPI -> IEEECigreAPIInputSignals.

apiNumIntStates	0..1	Integer	Size of internal integer array storage needed by this model for state variables. These are not represented in CIM, but the information could help identify different versions of this model. The EMT/RMS simulator manages this memory.
apiNumOutputPorts	0..1	Integer	The number of output ports expected by this model, which should match cardinality of the associated IEEECigreAPI -> IEEECigreAPIOutputSignals.
apiNumParameters	0..1	Integer	The number of input parameters expected by this model, which should match cardinality of the associated IEEECigreAPI -> IEEECigreAPIParameters.
IEEECigreAPIs	0..*	IEEECigreAPI	Minimum set of attributes required to use this model.

IEEECigreAPIInput

Emtiop

Connects the set of CIGRE TB 958 API input signals, for this instance, to points in the network model or to external references, like other controllers.

Native Members

kind	0..1	IEEECigreAPIInputKind	The type of input signal. If remoteInputSignal, supply the RemoteInputSignal association. The phase attribute is required for acTerminalVoltage, acCurrentVsc, and acCurrentGrid. If apiDefined, the model must be queried through its API for more information.
sensorRatio	0..1	Float	Ratio between measured quantity on the power network and signal quantity in the control system, e.g., a current transformer (CT) or voltage transformer (VT) ratio. Should be greater than or equal to 1.

Inherited Members

apiName	0..1	String	see IEEECigreAPISignal
apiParameterKind	0..1	IEEECigreAPIParameterKind	see IEEECigreAPISignal

apiSequenceNumber	0..1	Integer	see IEEECigreAPISignal
apiWidth	0..1	Integer	see IEEECigreAPISignal
multiplier	0..1	UnitMultiplier	see IEEECigreAPISignal
phase	0..1	SinglePhaseKind	see IEEECigreAPISignal
unit	0..1	UnitSymbol	see IEEECigreAPISignal
ConnectivityNode	0..1	ConnectivityNode	see IEEECigreAPISignal
DCNode	0..1	DCNode	see IEEECigreAPISignal
IEEECigreAPISignalInfo	0..1	IEEECigreAPISignalInfo	see IEEECigreAPISignal
ACDCTerminal	0..1	ACDCTerminal	see SignalDescriptor
DynamicsFunctionBlock	0..1	DynamicsFunctionBlock	see SignalDescriptor
mRID	0..1	String	see DetailedModelDescriptor

DetailedModelTypeDynamics	0..1	DetailedModelTypeDynamics	see DetailedModelDescriptor
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

IEEECigreAPIOutput

Emtiop

Connects the set of CIGRE TB 958 API output signals, for this instance, to points in the network model.

Native Members

kind	0..1	IEEECigreAPIOutputKind	The type of output signal. The phase attribute must be supplied with modulationIndex and vscVoltage. If apiDefined, obtain more information from the CIGRE TB 958 API.
scalingRatio	0..1	Float	Ratio between the controller output and the power system connection point. For example, a modulation index output could be multiplied by 50% of the DC link voltage of a converter to create a controlled voltage source connected to the AC

power network. If the DC link voltage is 1200 V, the scalingRatio should then be 600. May be any value, noting that a value of zero has no effect on the power network.

Inherited Members

apiName	0..1	String	see IEEECigreAPISignal
apiParameterKind	0..1	IEEECigreAPIParameterKind	see IEEECigreAPISignal
apiSequenceNumber	0..1	Integer	see IEEECigreAPISignal
apiWidth	0..1	Integer	see IEEECigreAPISignal
multiplier	0..1	UnitMultiplier	see IEEECigreAPISignal
phase	0..1	SinglePhaseKind	see IEEECigreAPISignal
unit	0..1	UnitSymbol	see IEEECigreAPISignal
ConnectivityNode	0..1	ConnectivityNode	see IEEECigreAPISignal

DCNode	0..1	DCNode	see IEEECigreAPISignal
IEEECigreAPISignalInfo	0..1	IEEECigreAPISignalInfo	see IEEECigreAPISignal
ACDCTerminal	0..1	ACDCTerminal	see SignalDescriptor
DynamicsFunctionBlock	0..1	DynamicsFunctionBlock	see SignalDescriptor
mRID	0..1	String	see DetailedModelDescriptor
DetailedModelTypeDynamics	0..1	DetailedModelTypeDynamics	see DetailedModelDescriptor
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

IEEECigreAPIParameter

Emtiop

A single value in the array of CIGRE TB 958 API input values. The meaning of this parameter is discoverable through the API for an implementation, like a DLL, and/or documentation provided with the model. This CIM class maintains only the essential parameter setting and location/size in the array of API inputs.

Native Members

apiParameterKind	0..1	IEEECigreAPIParameterKind	The C type of this parameter as expected by the CIGRE TB 958 API. This also determines the memory size of this parameter in the array of model inputs. It indicates whether the value attribute should be considered a string, integer, or floating point value.
apiSequenceNumber	0..1	Integer	The zero-based array index for this parameter, as expected in the CIGRE TB 958 API.
value	0..1	String	The parameter value, to be parsed from string format according to the apiParameterKind.
IEEECigreAPI	0..1	IEEECigreAPI	The API model instance associated with this parameter.
IEEECigreAPIParameterInfo	0..1	IEEECigreAPIParameterInfo	

Expanded set of attributes available from the CIGRE TB 958 API.

IEEECigreAPIParameterInfo

Emtiop

Supplemental information about this parameter from the CIGRE TB 958 API.

Native Members

apiDefaultValue	0..1	String	Default value internally assigned by the model implementation.
apiDescription	0..1	String	General description.
apiFixedValue	0..1	Boolean	True if the parameter can be changed any time during simulation, False if the parameter value must be set and time zero and not changed thereafter.
apiGroupName	0..1	String	A group name, if applicable.
apiMaxValue	0..1	String	

			Maximum value allowed, for numerical parameters.
apiMinValue	0..1	String	Minimum value allowed, for numerical parameters.
apiName	0..1	String	The name of this parameter, as returned by the CIGRE TB 958 API. If there is an inherited IdentifiedObject.name attribute, it may not necessarily match this name.
apiUnit	0..1	String	The parameter units expected by the CIGRE TB 958 API. This may not correspond to CIM units, so interpretation may be required.
IEEECigreAPIParameters	0..*	IEEECigreAPIParameter	Minimum set of attributes required to use this parameter.

IEEECigreAPISignalInfo

Emtiop

Supplemental information about the signal from the CIGRE TB 958 API.

Native Members

apiDescription	0..1	String	A description of the signal as written by the model's developer.
apiUnit	0..1	String	The signal units expected by the CIGRE TB 958 API. This may not correspond to CIM units, so interpretation may be required.
IEEECigreAPISignals	0..*	IEEECigreAPISignal	Minimum set of attributes required to use this signal.

LinearShuntCompensator

Wires

A linear shunt compensator has banks or sections with equal admittance values.

Native Members

bPerSection	1..1	Susceptance	Positive sequence shunt (charging) susceptance per section.

gPerSection	1..1	Conductance	Positive sequence shunt (charging) conductance per section.

Inherited Members

grounded	1..1	Boolean	see ShuntCompensator
maximumSections	1..1	Integer	see ShuntCompensator
nomU	1..1	Voltage	see ShuntCompensator
phaseConnection	0..1	PhaseShuntConnectionKind	see ShuntCompensator
sections	1..1	Float	see ShuntCompensator
BaseVoltage	0..1	BaseVoltage	see ConductingEquipment
inService	1..1	Boolean	see Equipment
EquipmentContainer	1..1	EquipmentContainer	see Equipment

mRID	1..1	String	see PowerSystemResource
name	1..1	String	see PowerSystemResource
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

LoadResponseCharacteristic

LoadModel

Models the characteristic response of the load demand due to changes in system conditions such as voltage and frequency. It is not related to demand response.

If LoadResponseCharacteristic.exponentModel is True, the exponential voltage or frequency dependent models are specified and used as to calculate active and reactive power components of the load model.

The equations to calculate active and reactive power components of the load model are internal to the power flow calculation, hence they use different quantities depending on the use case of the data exchange.

The equations for exponential voltage dependent load model injected power are:

pInjection= Pnominal* (Voltage/cim:BaseVoltage.nominalVoltage) ** cim:LoadResponseCharacteristic.pVoltageExponent

qInjection= Qnominal* (Voltage/cim:BaseVoltage.nominalVoltage) ** cim:LoadResponseCharacteristic.qVoltageExponent

pInjection = Pnominal* (Frequency/(Nominal frequency))**cim:LoadResponseCharacteristic.pFrequencyExponent

qInjection = Qnominal* (Frequency/(Nominal frequency))**cim:LoadResponseCharacteristic.qFrequencyExponent

Note that both voltage and frequency exponents could be used together so the full equation would be:

pInjection = Pnominal* (Voltage/(cim:BaseVoltage.nominalVoltage))**cim:LoadResponseCharacteristic.pVoltageExponent * (Frequency/(base frequency))**cim:LoadResponseCharacteristic.pFrequencyExponent

qInjection = Qnominal* (Voltage/(cim:BaseVoltage.nominalVoltage))**cim:LoadResponseCharacteristic.qVoltageExponent * (Frequency/(base frequency))**cim:LoadResponseCharacteristic.qFrequencyExponent

The voltage and frequency expressed in the equation are values obtained from solved power flow. Base voltage and base frequency are those derived from the connectivity of the static network model.

Where:

1) * means "multiply" and ** is "raised to the power of";

2) Pnominal and Qnominal represent the active power and reactive power at nominal voltage as any load described by the voltage exponential model shall be given at nominal voltage. This means that EnergyConsumer.p and EnergyConsumer.q are at nominal voltage.

3) After power flow is solved:

-pInjection and qInjection correspond to SvPowerflow.p and SvPowerflow.q respectively.

- Voltage corresponds to SvVoltage.v at the TopologicalNode where the load is connected.

Native Members

exponentModel	1..1	Boolean	Indicates the exponential voltage dependency model is to be used. If false, the coefficient model is to be used. The exponential voltage dependency model consist of the attributes: - pVoltageExponent - qVoltageExponent - pFrequencyExponent - qFrequencyExponent. The coefficient model consist of the attributes: - pConstantImpedance - pConstantCurrent - pConstantPower - qConstantImpedance - qConstantCurrent - qConstantPower.

			The sum of pConstantImpedance, pConstantCurrent and pConstantPower shall equal 1. The sum of qConstantImpedance, qConstantCurrent and qConstantPower shall equal 1.
pConstantCurrent	1..1	Float	Portion of active power load modelled as constant current.
pConstantImpedance	1..1	Float	Portion of active power load modelled as constant impedance.
pConstantPower	1..1	Float	Portion of active power load modelled as constant power.
pFrequencyExponent	1..1	Float	Exponent of per unit frequency effecting active power.
pVoltageExponent	1..1	Float	Exponent of per unit voltage effecting real power.
qConstantCurrent	1..1	Float	Portion of reactive power load modelled as constant current.

qConstantImpedance	1..1	Float	Portion of reactive power load modelled as constant impedance.
qConstantPower	1..1	Float	Portion of reactive power load modelled as constant power.
qFrequencyExponent	1..1	Float	Exponent of per unit frequency effecting reactive power.
qVoltageExponent	1..1	Float	Exponent of per unit voltage effecting reactive power.

Inherited Members

mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

MachineSaturationCurve

Emtiop

Use to define machine saturation with more than two points. xUnit is A (RMS current) and y1Unit is none (per-unit voltage).

Native Members

SynchronousMachineDetailed	0..1	SynchronousMachineDetailed	The synchronous machine this saturation characteristic applies to.

Inherited Members

mRID	1..1	String	see Curve
curveStyle	0..1	CurveStyle	see Curve
name	1..1	String	see Curve
xMultiplier	0..1	UnitMultiplier	see Curve
xUnit	0..1	UnitSymbol	see Curve
y1Multiplier	0..1	UnitMultiplier	see Curve
y1Unit	0..1	UnitSymbol	see Curve

mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

NthAmDynamicModel

Emtiop

Parameterized models from software libraries or user code that rely on documentation provided elsewhere, e.g., software documentation. The model is named within the domain of nameKind, e.g., ST6B for DYR or esst6b for DYD. Parameters are maintained by name and sequence number in the ParameterDescriptor class.

Native Members

closestStandardModel	0..1	String	Name of the closest match from Dynamics / StandardModels, if such a match exists.
modelKind	0..1	NthAmModelKind	Suggested application of this dynamic model.
nameKind	0..1	NthAmModelNameKind	
statusKind	0..1	NthAmModelStatusKind	

Inherited Members

NuclearGeneratingUnit

Production

A nuclear generating unit.

Inherited Members

maxOperatingP	1..1	ActivePower	see GeneratingUnit
minOperatingP	1..1	ActivePower	see GeneratingUnit
inService	1..1	Boolean	see Equipment
EquipmentContainer	1..1	EquipmentContainer	see Equipment
mRID	1..1	String	see PowerSystemResource
name	1..1	String	see PowerSystemResource
mRID	0..1	String	see IdentifiedObject

name	0..1	String	see IdentifiedObject

OperationalLimitSet

OperationalLimits

A set of limits associated with equipment. Sets of limits might apply to a specific temperature, or season for example. A set of limits may contain different severities of limit levels that would apply to the same equipment. The set may contain limits of different types such as apparent power and current limits or high and low voltage limits that are logically applied together as a set.

Native Members

mRID	1..1	String	Master resource identifier issued by a model authority. The mRID is unique within an exchange context. Global uniqueness is easily achieved by using a UUID, as specified in IETF RFC 4122, for the mRID. The use of UUID is strongly recommended. For CIMXML data files in RDF syntax conforming to IEC 61970-552, the mRID is mapped to rdf:ID or rdf:about attributes that identify CIM object elements.

name	1..1	String	The name is any free human readable and possibly non unique text naming the object.
Terminal	1..1	ACDCTerminal	The terminal where the operational limit set apply.

OperationalLimitType

OperationalLimits

The operational meaning of a category of limits.

Native Members

mRID	1..1	String	Master resource identifier issued by a model authority. The mRID is unique within an exchange context. Global uniqueness is easily achieved by using a UUID, as specified in IETF RFC 4122, for the mRID. The use of UUID is strongly recommended. For CIMXML data files in RDF syntax conforming to IEC 61970-552, the mRID is mapped to rdf:ID or rdf:about

			attributes that identify CIM object elements.
acceptableDuration	0..1	Seconds	The nominal acceptable duration of the limit. Limits are commonly expressed in terms of the time limit for which the limit is normally acceptable. The actual acceptable duration of a specific limit may depend on other local factors such as temperature or wind speed. The attribute has meaning only if the flag isInfiniteDuration is set to false, hence it shall not be exchanged when isInfiniteDuration is set to true.
direction	0..1	OperationalLimitDirectionKind	The direction of the limit.
isInfiniteDuration	0..1	Boolean	Defines if the operational limit type has infinite duration. If true, the limit has infinite duration. If false, the limit has definite duration which is defined by the attribute acceptableDuration.

name	1..1	String	The name is any free human readable and possibly non unique text naming the object.

ParameterDescriptor

DetailedModelDescription

Supports definition of one or more parameters of several different datatypes for use by the detailed model. It describes the parameters used in the equations of the detailed model.

The name of the parameter shall be the same as the name used in the equations of the detailed model.

Native Members

engineeringUnit	0..1	String	The engineering unit of the value.
sequenceNumber	0..1	Integer	Sequence number of the parameter among the set of parameters associated with the related proprietary user-defined model.
typicalValue	0..1	String	Typical value for the parameter. The datatype is as specified in attribute valueXSDdatatype.

Inherited Members

mRID	0..1	String	see DetailedModelDescriptor
DetailedModelTypeDynamics	0..1	DetailedModelTypeDynamics	see DetailedModelDescriptor
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

ParameterValue

DetailedModelDescription

Provides the value of a given parameter of a detailed model dynamics.

Native Members

value	0..1	String	The value of the parameter.
DetailedModelDynamics	0..1	DetailedModelDynamics	The detailed model to which this parameter value applies.

ParameterDescriptor	0..1	ParameterDescriptor	The parameter descriptor that has this value.

PhaseTapChangerLinear

Wires

Describes a tap changer with a linear relation between the tap step and the phase angle difference across the transformer. This is a mathematical model that is an approximation of a real phase tap changer.

The phase angle is computed as stepPhaseShiftIncrement times the tap position.

The voltage magnitude of both sides is the same.

Native Members

stepPhaseShiftIncrement	0..1	AngleDegrees	Phase shift per step position. A positive value indicates a positive angle variation from the Terminal at the PowerTransformerEnd, where the TapChanger is located, into the transformer. The actual phase shift increment might be more accurately computed from the symmetrical or asymmetrical models or a tap step table lookup if those are available.

xMax	0..1	Reactance	The reactance depends on the tap position according to a "u" shaped curve. The maximum reactance (xMax) appears at the low and high tap positions. Depending on the "u" curve the attribute can be either higher or lower than PowerTransformerEnd.x.

Inherited Members

TransformerEnd	0..1	TransformerEnd	see PhaseTapChanger
highStep	0..1	Integer	see TapChanger
lowStep	0..1	Integer	see TapChanger
neutralStep	0..1	Integer	see TapChanger
neutralU	0..1	Voltage	see TapChanger
normalStep	0..1	Integer	see TapChanger

step	1..1	Float	see TapChanger
mRID	1..1	String	see PowerSystemResource
name	1..1	String	see PowerSystemResource
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

PhotoVoltaicUnit

Production

A photovoltaic device or an aggregation of such devices.

Inherited Members

maxP	1..1	ActivePower	see PowerElectronicsUnit
minP	1..1	ActivePower	see PowerElectronicsUnit

PowerElectronicsConnection	1..1	PowerElectronicsConnection	see PowerElectronicsUnit
inService	1..1	Boolean	see Equipment
EquipmentContainer	1..1	EquipmentContainer	see Equipment
mRID	1..1	String	see PowerSystemResource
name	1..1	String	see PowerSystemResource
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

PowerElectronicsConnection

Wires

A connection to the AC network for energy production or consumption that uses power electronics rather than rotating machines.

Native Members

maxIFault	1..1	PU	Maximum fault current this device will contribute, in per-unit of rated current, before the converter protection will trip or bypass.
maxQ	1..1	ReactivePower	Maximum reactive power limit. This is the maximum (nameplate) limit for the unit.
minQ	1..1	ReactivePower	Minimum reactive power limit for the unit. This is the minimum (nameplate) limit for the unit.
p	1..1	ActivePower	Active power injection. Load sign convention is used, i.e. positive sign means flow out from a node. Starting value for a steady state solution.
q	1..1	ReactivePower	Reactive power injection. Load sign convention is used, i.e. positive sign means flow out from a node. Starting value for a steady state solution.

ratedS	1..1	ApparentPower	Nameplate apparent power rating for the unit. The attribute shall have a positive value.
ratedU	1..1	Voltage	Rated voltage (nameplate data, Ur in IEC 60909-0). It is primarily used for short circuit data exchange according to IEC 60909. The attribute shall be a positive value.

Inherited Members

BaseVoltage	0..1	BaseVoltage	see ConductingEquipment
inService	1..1	Boolean	see Equipment
EquipmentContainer	1..1	EquipmentContainer	see Equipment
mRID	1..1	String	see PowerSystemResource

name	1..1	String	see PowerSystemResource
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

PowerElectronicsConnectionDCTerminal

Emtiop

A DC connection point at the converter, which is also connected on the AC side as any other AC ConductingEquipment. This special terminal is separate from the regular DCTerminal to restrict the connection, such that no other DC conducting equipment can be connected to the AC side.

Native Members

polarity	0..1	DCTerminalPolarityKind	Use to indicated positive or negative polarity on the DC side.
PowerElectronicsConnection	0..1	PowerElectronicsConnection	The PowerElectronicsConnection for this terminal.

Inherited Members

DCNode	1..1	DCNode	see DCBaseTerminal
sequenceNumber	1..1	Integer	see ACDCTerminal
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

PowerElectronicsWindUnit

Production

A wind generating unit that connects to the AC network with power electronics rather than rotating machines or an aggregation of such units.

Inherited Members

maxP	1..1	ActivePower	see PowerElectronicsUnit
minP	1..1	ActivePower	see PowerElectronicsUnit

PowerElectronicsConnection	1..1	PowerElectronicsConnection	see PowerElectronicsUnit
inService	1..1	Boolean	see Equipment
EquipmentContainer	1..1	EquipmentContainer	see Equipment
mRID	1..1	String	see PowerSystemResource
name	1..1	String	see PowerSystemResource
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

PowerTransformer

Wires

An electrical device consisting of two or more coupled windings, with or without a magnetic core, for introducing mutual coupling between electric circuits. Transformers can be used to control voltage and phase shift (active power flow).

A power transformer may be composed of separate transformer tanks that need not be identical.

A power transformer can be modelled with or without tanks and is intended for use in both balanced and unbalanced representations. A power transformer typically has two terminals, but may have one (grounding), three or more terminals.

The inherited association ConductingEquipment.BaseVoltage should not be used. The association from TransformerEnd to BaseVoltage should be used instead.

Native Members

vectorGroup	1..1	String	Vector group of the transformer for protective relaying, e.g., Dyn1. For unbalanced transformers, this may not be simply determined from the constituent winding connections and phase angle displacements. The vectorGroup string consists of the following components in the order listed: high voltage winding connection, mid voltage winding connection (for three winding transformers), phase displacement clock number from 0 to 11, low voltage winding connection phase displacement clock number from 0 to 11. The winding connections are D (delta), Y (wye), YN (wye with neutral), Z (zigzag), ZN (zigzag with neutral), A (auto transformer). Upper case means the high voltage, lower case mid or low. The high voltage winding always has

			clock position 0 and is not included in the vector group string. Some examples: YNy0 (two winding wye to wye with no phase displacement), YNd11 (two winding wye to delta with 330 degrees phase displacement), YNyn0d5 (three winding transformer wye with neutral high voltage, wye with neutral mid voltage and no phase displacement, delta low voltage with 150 degrees displacement). Phase displacement is defined as the angular difference between the phasors representing the voltages between the neutral point (real or imaginary) and the corresponding terminals of two windings, a positive sequence voltage system being applied to the high-voltage terminals, following each other in alphabetical sequence if they are lettered, or in numerical sequence if they are numbered: the phasors are assumed to rotate in a counter-clockwise sense.

Inherited Members

BaseVoltage	0..1	BaseVoltage	see ConductingEquipment

inService	1..1	Boolean	see Equipment
EquipmentContainer	1..1	EquipmentContainer	see Equipment
mRID	1..1	String	see PowerSystemResource
name	1..1	String	see PowerSystemResource
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

PowerTransformerEnd

Wires

A PowerTransformerEnd is associated with each Terminal of a PowerTransformer.

The impedance values r, r0, x, and x0 of a PowerTransformerEnd represents a star equivalent as follows.

1) two PowerTransformerEnd-s shall be defined for a two Terminal PowerTransformer even if the two PowerTransformerEnd-s have the same rated voltage. The high voltage PowerTransformerEnd (TransformerEnd.endNumber=1) is the one used to exchange resistances (r, r0) and reactances (x, x0) of the PowerTransformer while the low voltage PowerTransformerEnd (TransformerEnd.endNumber=2) shall have zero impedance values.

2) for a three Terminal PowerTransformer the three PowerTransformerEnds represent a star equivalent with each leg in the star represented by r, r0, x, and x0 values.

3) For a three Terminal transformer each PowerTransformerEnd shall have g, g0, b and b0 values corresponding to the no load losses distributed on the three PowerTransformerEnds. The total no load loss shunt impedances may also be placed at one of the PowerTransformerEnds, preferably the end numbered 1, having the shunt values on end 1. This is the preferred way.

4) for a PowerTransformer with more than three Terminals the PowerTransformerEnd impedance values cannot be used. Instead use the TransformerMeshImpedance or split the transformer into multiple PowerTransformers.

Each PowerTransformerEnd must be contained by a PowerTransformer. Because a PowerTransformerEnd (or any other object) can not be contained by more than one parent, a PowerTransformerEnd can not have an association to an EquipmentContainer (Substation, VoltageLevel, etc).

Native Members

connectionKind	1..1	WindingConnection	Kind of connection.
phaseAngleClock	1..1	Integer	Terminal voltage phase angle displacement where 360 degrees are represented with clock hours. The valid values are 0 to 11. For example, for the secondary side end of a transformer with vector group code of 'Dyn11', specify the connection kind as wye with neutral and specify the phase angle of the clock as 11. The clock value of the transformer end number specified as 1, is assumed to be zero. Note the transformer end number is not

			assumed to be the same as the terminal sequence number.
ratedS	1..1	ApparentPower	Normal apparent power rating. The attribute shall be a positive value. For a two-winding transformer the values for the high and low voltage sides shall be identical.
ratedU	1..1	Voltage	Rated voltage: phase-phase for three-phase windings, and either phase-phase or phase-neutral for single-phase windings. A high voltage side, as given by TransformerEnd.endNumber, shall have a ratedU that is greater than or equal to ratedU for the lower voltage sides. The attribute shall be a positive value.
PowerTransformer	1..1	PowerTransformer	The power transformer of this power transformer end.

Inherited Members

mRID	1..1	String	see TransformerEnd
endNumber	1..1	Integer	see TransformerEnd
grounded	1..1	Boolean	see TransformerEnd
name	1..1	String	see TransformerEnd
rground	1..1	Resistance	see TransformerEnd
xground	1..1	Reactance	see TransformerEnd
BaseVoltage	1..1	BaseVoltage	see TransformerEnd
Terminal	1..1	Terminal	see TransformerEnd
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

RatioTapChanger

A tap changer that changes the voltage ratio impacting the voltage magnitude but not the phase angle across the transformer.

Angle sign convention (general): Positive value indicates a positive phase shift from the winding where the tap is located to the other winding (for a two-winding transformer).

Native Members

stepVoltageIncrement	1..1	PerCent	Tap step increment, in per cent of rated voltage of the power transformer end, per step position. When the increment is negative, the voltage decreases when the tap step increases.
TransformerEnd	1..1	TransformerEnd	Transformer end to which this ratio tap changer belongs.

Inherited Members

highStep	0..1	Integer	see TapChanger

lowStep	0..1	Integer	see TapChanger
neutralStep	0..1	Integer	see TapChanger
neutralU	0..1	Voltage	see TapChanger
normalStep	0..1	Integer	see TapChanger
step	1..1	Float	see TapChanger
mRID	1..1	String	see PowerSystemResource
name	1..1	String	see PowerSystemResource
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

RotatingMachinePlant

Emtiop

A conventional generating plant, e.g., SynchronousMachine with associated controls and GeneratingUnit, a PowerTransformer, and a DisconnectingCircuitBreaker for thermal and hydro plants.

Inherited Members

ACPointOfCommonCoupling	0..1	ACPointOfCommonCoupling	see ConnectedFacility
Equipments	0..unbounded	Equipment	see ConnectedFacility
mRID	1..1	String	see PowerSystemResource
name	1..1	String	see PowerSystemResource
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

SeriesCompensator

Wires

A Series Compensator is a series capacitor or reactor or an AC transmission line without charging susceptance. It is a two terminal device.

Native Members

r	1..1	Resistance	Positive sequence resistance.
r0	1..1	Resistance	Zero sequence resistance.
x	1..1	Reactance	Positive sequence reactance.
x0	1..1	Reactance	Zero sequence reactance.

Inherited Members

BaseVoltage	0..1	BaseVoltage	see ConductingEquipment
inService	1..1	Boolean	see Equipment
EquipmentContainer	1..1	EquipmentContainer	see Equipment
mRID	1..1	String	see PowerSystemResource
name	1..1	String	see PowerSystemResource

mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

StaticVarCompensator

FACTS

A facility for providing variable and controllable shunt reactive power. The SVC typically consists of a stepdown transformer, filter, thyristor-controlled reactor, and thyristor-switched capacitor arms.

The SVC may operate in fixed MVar output mode or in voltage control mode. When in voltage control mode, the output of the SVC will be proportional to the deviation of voltage at the controlled bus from the voltage setpoint. The SVC characteristic slope defines the proportion. If the voltage at the controlled bus is equal to the voltage setpoint, the SVC MVar output is zero.

Native Members

capacitiveRating	0..1	Reactance	Capacitive reactance at maximum capacitive reactive power. Shall always be positive.
inductiveRating	0..1	Reactance	Inductive reactance at maximum inductive reactive power. Shall always be negative.

q	0..1	ReactivePower	Reactive power injection. Load sign convention is used, i.e. positive sign means flow out from a node. Starting value for a steady state solution.
slope	0..1	VoltagePerReactivePower	The characteristics slope of an SVC defines how the reactive power output changes in proportion to the difference between the regulated bus voltage and the voltage setpoint. The attribute shall be a positive value or zero.
sVCControlMode	0..1	SVCControlMode	SVC control mode.
voltageSetPoint	0..1	Voltage	The reactive power output of the SVC is proportional to the difference between the voltage at the regulated bus and the voltage setpoint. When the regulated bus voltage is equal to the voltage setpoint, the reactive power output is zero.

Inherited Members

BaseVoltage	0..1	BaseVoltage	see ConductingEquipment
inService	1..1	Boolean	see Equipment
EquipmentContainer	1..1	EquipmentContainer	see Equipment
mRID	1..1	String	see PowerSystemResource
name	1..1	String	see PowerSystemResource
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

SvDCPowerFlow

StateVariables

State variable for power flow. Load convention is used for flow direction. This means flow out from the DCTopologicalNode into the equipment is positive.

Native Members

p	0..1	ActivePower	The active power flow. Load sign convention is used, i.e. positive sign means flow out from a DCTopologicalNode (bus) into the conducting equipment.
DCTerminal	0..1	DCTerminal	The DC terminal associated with the DC power flow state variable.

Inherited Members

SvDCVoltage

StateVariables

State variable for direct current voltage.

Native Members

v	0..1	Voltage	State variable for direct current voltage.
DCTopologicalNode	0..1	DCTopologicalNode	The DC topological node associated with the DC voltage state.

Inherited Members

SvPowerFlow

StateVariables

State variable for power flow. Load convention is used for flow direction. This means flow out from the TopologicalNode into the equipment is positive.

Native Members

p	0..1	ActivePower	The active power flow. Load sign convention is used, i.e. positive sign means flow out from a TopologicalNode (bus) into the conducting equipment.
q	0..1	ReactivePower	The reactive power flow. Load sign convention is used, i.e. positive sign means flow out from a TopologicalNode (bus) into the conducting equipment.
Terminal	0..1	Terminal	The terminal associated with the power flow state variable.

Inherited Members

SvShuntCompensatorSections

StateVariables

State variable for the number of sections in service for a shunt compensator.

Native Members

phase	0..1	SinglePhaseKind	The terminal phase at which the connection is applied. If missing, the injection is assumed to be balanced among non-neutral phases.
sections	0..1	Float	The number of sections in service as a continuous variable. The attribute shall be a positive value or zero. To get integer value scale with ShuntCompensator.bPerSection.
ShuntCompensator	0..1	ShuntCompensator	The shunt compensator for which the state applies.

Inherited Members

SvStatus

State variable for status.

Native Members

inService	0..1	Boolean	The in service status as a result of topology processing. It indicates if the equipment is considered as energized by the power flow. It reflects if the equipment is connected within a solvable island. It does not necessarily reflect whether or not the island was solved by the power flow.
phase	0..1	SinglePhaseKind	The individual phase status. If the attribute is unspecified, then three phase model is assumed.
ConductingEquipment	0..1	ConductingEquipment	The conducting equipment associated with the status state variable.

Inherited Members

SvSwitch

StateVariables

State variable for switch.

Native Members

open	0..1	Boolean	The attribute tells if the computed state of the switch is considered open.
phase	0..1	SinglePhaseKind	The terminal phase at which the connection is applied. If missing, the injection is assumed to be balanced among non-neutral phases.
Switch	0..1	Switch	The switch associated with the switch state.

Inherited Members

SvTapStep

StateVariables

State variable for transformer tap step.

Native Members

position	0..1	Float	The floating point tap position. This is not the tap ratio, but rather the tap step position as defined by the related tap changer model and normally is constrained to be within the range of minimum and maximum tap positions.
TapChanger	0..1	TapChanger	The tap changer associated with the tap step state.

Inherited Members

SvVoltage

StateVariables

State variable for voltage.

Native Members

angle	0..1	AngleDegrees	The voltage angle of the topological node complex voltage with respect to system reference.

v	0..1	Voltage	The voltage magnitude at the topological node. The attribute shall be a positive value.
TopologicalNode	0..1	TopologicalNode	The topological node associated with the voltage state.

Inherited Members

SynchronousMachine

Wires

An electromechanical device that operates with shaft rotating synchronously with the network. It is a single machine operating either as a generator or synchronous condenser or pump.

Native Members

earthing	1..1	Boolean	Indicates whether or not the generator is earthed. Used for short circuit data exchange according to IEC 60909.
earthingStarPointR	1..1	Resistance	Generator star point earthing resistance (Re). Used for short circuit data

			exchange according to IEC 60909.
earthingStarPointX	1..1	Reactance	Generator star point earthing reactance (Xe). Used for short circuit data exchange according to IEC 60909.
maxQ	1..1	ReactivePower	Maximum reactive power limit. This is the maximum (nameplate) limit for the unit.
minQ	1..1	ReactivePower	Minimum reactive power limit for the unit.
operatingMode	1..1	SynchronousMachineOperatingMode	Current mode of operation.
type	1..1	SynchronousMachineKind	Modes that this synchronous machine can operate in.

Inherited Members

p	1..1	ActivePower	see RotatingMachine
q	1..1	ReactivePower	see RotatingMachine
ratedS	1..1	ApparentPower	see RotatingMachine
ratedU	1..1	Voltage	see RotatingMachine
GeneratingUnit	1..1	GeneratingUnit	see RotatingMachine
BaseVoltage	0..1	BaseVoltage	see ConductingEquipment
inService	1..1	Boolean	see Equipment
EquipmentContainer	1..1	EquipmentContainer	see Equipment
mRID	1..1	String	see PowerSystemResource
name	1..1	String	see PowerSystemResource

mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

SynchronousMachineSimplified

SynchronousMachineDynamics

The simplified model represents a synchronous generator as a constant internal voltage behind an impedance (*Rs* + *jXp*) as shown in the Simplified diagram.

Since internal voltage is held constant, there is no *Efd* input and any excitation system model will be ignored. There is also no *Ifd* output.

This model should not be used for representing a real generator except, perhaps, small generators whose response is insignificant.

The parameters used for the simplified model include:

- RotatingMachineDynamics.damping (*D*);

- RotatingMachineDynamics.inertia (*H*);

- RotatingMachineDynamics.statorLeakageReactance (used to exchange *jXp* for SynchronousMachineSimplified);

- RotatingMachineDynamics.statorResistance (*Rs*).

Inherited Members

SynchronousMachine	0..1	SynchronousMachine	see SynchronousMachineDynamics
damping	1..1	Float	see RotatingMachineDynamics
inertia	1..1	Seconds	see RotatingMachineDynamics
statorLeakageReactance	1..1	PU	see RotatingMachineDynamics
statorResistance	1..1	PU	see RotatingMachineDynamics
mRID	1..1	String	see DynamicsFunctionBlock
enabled	1..1	Boolean	see DynamicsFunctionBlock
name	1..1	String	see DynamicsFunctionBlock
mRID	0..1	String	see IdentifiedObject

name	0..1	String	see IdentifiedObject

SynchronousMachineTimeConstantReactance SynchronousMachineDynamics

Synchronous machine detailed modelling types are defined by the combination of the attributes SynchronousMachineTimeConstantReactance.modelType and SynchronousMachineTimeConstantReactance.rotorType.

Parameter details:

1. The â€œpâ€� in the time-related attribute names is a substitution for a â€œprimeâ€� in the usual parameter notation, e.g. tpdo refers to *T'do*.

Native Members

ks	1..1	Float	Saturation loading correction factor (*Ks*) (>= 0). Used only by type J model. Typical value = 0.
modelType	1..1	SynchronousMachineModelKind	Type of synchronous machine model used in dynamic simulatio applications.
rotorType	1..1	RotorKind	Type of rotor on physical machine.
tc	1..1	Seconds	

			Damping time constant for â€œCanayâ€� reactance (>= 0). Typical value = 0.
tpdo	1..1	Seconds	Direct-axis transient rotor time constant (*T'do*) (> SynchronousMachineTimeConstantReactance.tppdo). Typical value = 5.
tppdo	1..1	Seconds	Direct-axis subtransient rotor time constant (*T''do*) (> 0). Typical value = 0,03.
tppqo	1..1	Seconds	Quadrature-axis subtransient rotor time constant (*T''qo*) (> 0). Typical value = 0,03.
tpqo	1..1	Seconds	Quadrature-axis transient rotor time constant (*T'qo*) (> SynchronousMachineTimeConstantReactance.tppqo). Typical value = 0,5.
xDirectSubtrans	1..1	PU	Direct-axis subtransient reactance (unsaturated) (*X''d*) (> RotatingMachineDynamics.statorLeakageReactance). Typical value = 0,2.
xDirectSync	1..1	PU	Direct-axis synchronous reactance (*Xd*) (>= SynchronousMachineTimeConstantReactance.xDirectTrans). The quotient of a sustained value of that AC component of armature voltage that is produced by the total direct-axis flux due to direct-axis armature current and the value of the AC

			component of this current, the machine running at rated speed. Typical value = 1,8.
xDirectTrans	1..1	PU	Direct-axis transient reactance (unsaturated) (*X'd*) (>= SynchronousMachineTimeConstantReactance.xDirectSubtrans). Typical value = 0,5.
xQuadSubtrans	1..1	PU	Quadrature-axis subtransient reactance (*X''q*) (> RotatingMachineDynamics.statorLeakageReactance). Typical value = 0,2.
xQuadSync	1..1	PU	Quadrature-axis synchronous reactance (*Xq*) (>= SynchronousMachineTimeConstantReactance.xQuadTrans). The ratio of the component of reactive armature voltage, due to the quadrature-axis component of armature current, to this component of current, under steady state conditions and at rated frequency. Typical value = 1,6.
xQuadTrans	1..1	PU	Quadrature-axis transient reactance (*X'q*) (>= SynchronousMachineTimeConstantReactance.xQuadSubtrans). Typical value = 0,3.

Inherited Members

efdBaseRatio	1..1	Float	see SynchronousMachineDetailed

ifdBaseType	1..1	IfdBaseKind	see SynchronousMachineDetailed
saturationFactor	1..1	Float	see SynchronousMachineDetailed
saturationFactor120	1..1	Float	see SynchronousMachineDetailed
saturationFactor120QAxis	1..1	Float	see SynchronousMachineDetailed
saturationFactorQAxis	1..1	Float	see SynchronousMachineDetailed
SynchronousMachine	0..1	SynchronousMachine	see SynchronousMachineDynamics
damping	1..1	Float	see RotatingMachineDynamics
inertia	1..1	Seconds	see RotatingMachineDynamics
statorLeakageReactance	1..1	PU	see RotatingMachineDynamics

statorResistance	1..1	PU	see RotatingMachineDynamics
mRID	1..1	String	see DynamicsFunctionBlock
enabled	1..1	Boolean	see DynamicsFunctionBlock
name	1..1	String	see DynamicsFunctionBlock
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Terminal

Core

An AC electrical connection point to a piece of conducting equipment. Terminals are connected at physical connection points called connectivity nodes.

Native Members

ConductingEquipment	1..1	ConductingEquipment	The conducting equipment of the terminal. Conducting equipment have terminals that may be

			connected to other conducting equipment terminals via connectivity nodes or topological nodes.
ConnectivityNode	1..1	ConnectivityNode	The connectivity node to which this terminal connects with zero impedance.

Inherited Members

sequenceNumber	1..1	Integer	see ACDCTerminal
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

TextDiagramObject

DiagramLayout

A diagram object for placing free-text or text derived from an associated domain object.

Native Members

text	1..1	String	The text that is displayed by this text diagram object.

Inherited Members

mRID	1..1	String	see DiagramObject
drawingOrder	1..1	Integer	see DiagramObject
isPolygon	1..1	Boolean	see DiagramObject
name	1..1	String	see DiagramObject
IdentifiedObject	1..1	IdentifiedObject	see DiagramObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

ThermalGeneratingUnit

Production

A generating unit whose prime mover could be a steam turbine, combustion turbine, or diesel engine.

Inherited Members

maxOperatingP	1..1	ActivePower	see GeneratingUnit
minOperatingP	1..1	ActivePower	see GeneratingUnit
inService	1..1	Boolean	see Equipment
EquipmentContainer	1..1	EquipmentContainer	see Equipment
mRID	1..1	String	see PowerSystemResource
name	1..1	String	see PowerSystemResource
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

TopologicalNode

Topology

For a detailed substation model a topological node is a set of connectivity nodes that, in the current network state, are connected together through any type of closed switches, including jumpers. Topological nodes change as the current network state changes (i.e., switches, breakers, etc. change state).

For a planning model, switch statuses are not used to form topological nodes. Instead they are manually created or deleted in a model builder tool. Topological nodes maintained this way are also called "busses".

Inherited Members

mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

TransformerCoreAdmittance

Wires

The transformer core admittance. Used to specify the core admittance of a transformer in a manner that can be shared among power transformers.

Native Members

mRID	1..1	String	Master resource identifier issued by a model authority. The mRID is unique within an exchange context.

			Global uniqueness is easily achieved by using a UUID, as specified in IETF RFC 4122, for the mRID. The use of UUID is strongly recommended. For CIMXML data files in RDF syntax conforming to IEC 61970-552, the mRID is mapped to rdf:ID or rdf:about attributes that identify CIM object elements.
b	1..1	Susceptance	Magnetizing branch susceptance (B mag). The value can be positive or negative.
b0	1..1	Susceptance	Zero sequence magnetizing branch susceptance.
g	1..1	Conductance	Magnetizing branch conductance (G mag).
g0	1..1	Conductance	Zero sequence magnetizing branch conductance.
name	1..1	String	The name is any free human readable and possibly non unique text naming the object.

TransformerEnd	1..1	TransformerEnd	All transformer ends having this core admittance.

Inherited Members

mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

TransformerMeshImpedance

Wires

Transformer mesh impedance (Delta-model) between transformer ends.

The typical case is that this class describes the impedance between two transformer ends pair-wise, i.e. the cardinalities at both transformer end associations are 1. However, in cases where two or more transformer ends are modelled the cardinalities are larger than 1.

Native Members

mRID	1..1	String	Master resource identifier issued by a model authority. The mRID is unique within an exchange context. Global uniqueness is easily achieved by using a UUID, as specified in IETF RFC 4122, for the mRID. The

			use of UUID is strongly recommended. For CIMXML data files in RDF syntax conforming to IEC 61970-552, the mRID is mapped to rdf:ID or rdf:about attributes that identify CIM object elements.
name	1..1	String	The name is any free human readable and possibly non unique text naming the object.
r	1..1	Resistance	Resistance between the 'from' and the 'to' end, seen from the 'from' end.
r0	1..1	Resistance	Zero-sequence resistance between the 'from' and the 'to' end, seen from the 'from' end.
x	1..1	Reactance	Reactance between the 'from' and the 'to' end, seen from the 'from' end.
x0	1..1	Reactance	Zero-sequence reactance between the 'from' and the 'to' end, seen from the 'from' end.

FromTransformerEnd	1..1	TransformerEnd	From end this mesh impedance is connected to. It determines the voltage reference.
ToTransformerEnd	1..*	TransformerEnd	All transformer ends this mesh impedance is connected to.

Inherited Members

mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

TransformerSaturationCurve

Emtiop

Represents a piecewise linear transformer saturation characteristic. It's connected to the same TransformerEnd as the TransformerCoreAdmittance, which is typically the lowest voltage winding other than a tertiary. The attached Curve is a piecewise linear magnetization characteristic, plotted as core flux linkage vs. magnetizing current. This replaces the linear magnetizing characteristic defined by TransformerCoreAdmittance.b and TransformerCoreAdmittance.b0. The TransformerSaturation characteristic does not include hysteresis, i.e., the origin point [0,0] is implied. The TransformerCoreAdmittance.g and TransformerCoreAdmittance.g0 should still be used to model core losses.

This data is of most interest to electromagnetic transient analysis, which typically uses SI units without multipliers.

xMultiplier inherited attribute should be UnitMultiplier.none

xUnit inherited attribute should be UnitSymbol.A

y1Multiplier inherited attribute should be UnitMultiplier.none

y1Unit inherited attribute should be UnitSymbol.Vs

xvalue in associated CurveData should be magnetizing current in peak A (not RMS), referenced to the TransformerEnd associated through TransformerCoreAdmittance. Do not enter the origin point [0,0].

y1value in associated CurveData should be core flux linkage in peak Vs (not RMS), referenced to the TransformerEnd associated through TransformerCoreAdmittance. Do not enter the origin point [0,0].

Native Members

TransformerCoreAdmittance	0..1	TransformerCoreAdmittance

Inherited Members

mRID	1..1	String	see Curve
curveStyle	0..1	CurveStyle	see Curve
name	1..1	String	see Curve
xMultiplier	0..1	UnitMultiplier	see Curve

xUnit	0..1	UnitSymbol	see Curve
y1Multiplier	0..1	UnitMultiplier	see Curve
y1Unit	0..1	UnitSymbol	see Curve
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

VsConverter

DC

DC side of the voltage source converter (VSC).

Native Members

delta	0..1	AngleDegrees	Angle between VsConverter.uv and ACDCConverter.uc. It is converter's state variable used in power flow. The attribute shall be a positive value or zero.

droop	0..1	PU	Droop constant. The pu value is obtained as D [kV/MW] * Sb / Ubdc. The attribute shall be a positive value.
droopCompensation	0..1	Resistance	Compensation constant. Used to compensate for voltage drop when controlling voltage at a distant bus. The attribute shall be a positive value.
maxModulationIndex	0..1	Float	The maximum quotient between the AC converter voltage (Uc) and DC voltage (Ud). A factor typically less than 1. It is converter's configuration data used in power flow.
maxValveCurrent	0..1	CurrentFlow	The maximum current through a valve. It is converter's configuration data.
pPccControl	0..1	VsPpccControlKind	Kind of control of real power and/or DC voltage.
qPccControl	0..1	VsQpccControlKind	Kind of reactive power control.

qShare	0..1	PerCent	Reactive power sharing factor among parallel converters on Uac control. The attribute shall be a positive value or zero.
targetPhasePcc	0..1	AngleDegrees	Phase target at AC side, at point of common coupling. The attribute shall be a positive value.
targetPowerFactorPcc	0..1	Float	Power factor target at the AC side, at point of common coupling. The attribute shall be a positive value.
targetPWMfactor	0..1	Float	Magnitude of pulse-modulation factor. The attribute shall be a positive value.
targetQpcc	0..1	ReactivePower	Reactive power injection target in AC grid, at point of common coupling. Load sign convention is used, i.e. positive sign means flow out from a node.
targetUpcc	0..1	Voltage	Voltage target in AC grid, at point of common coupling. The attribute shall be a positive value.

uv	0..1	Voltage	Line-to-line voltage on the valve side of the converter transformer. It is converter's state variable, result from power flow. The attribute shall be a positive value.

Inherited Members

baseS	0..1	ApparentPower	see ACDCConverter
idc	0..1	CurrentFlow	see ACDCConverter
idleLoss	0..1	ActivePower	see ACDCConverter
maxP	0..1	ActivePower	see ACDCConverter
maxUdc	0..1	Voltage	see ACDCConverter
minP	0..1	ActivePower	see ACDCConverter
minUdc	0..1	Voltage	see ACDCConverter

numberOfValves	0..1	Integer	see ACDCConverter
p	0..1	ActivePower	see ACDCConverter
poleLossP	0..1	ActivePower	see ACDCConverter
q	0..1	ReactivePower	see ACDCConverter
ratedUdc	0..1	Voltage	see ACDCConverter
resistiveLoss	0..1	Resistance	see ACDCConverter
switchingLoss	0..1	ActivePowerPerCurrentFlow	see ACDCConverter
targetPpcc	0..1	ActivePower	see ACDCConverter
targetUdc	0..1	Voltage	see ACDCConverter
uc	0..1	Voltage	see ACDCConverter
udc	0..1	Voltage	see ACDCConverter

valveU0	0..1	Voltage	see ACDCConverter
BaseVoltage	0..1	BaseVoltage	see ConductingEquipment
inService	1..1	Boolean	see Equipment
EquipmentContainer	1..1	EquipmentContainer	see Equipment
mRID	1..1	String	see PowerSystemResource
name	1..1	String	see PowerSystemResource
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Abstract Classes

ACDCConverter

DC

A unit with valves for three phases, together with unit control equipment, essential protective and switching devices, DC storage capacitors, phase reactors and auxiliaries, if any, used for conversion.

Native Members

baseS	0..1	ApparentPower	Base apparent power of the converter pole. The attribute shall be a positive value.
idc	0..1	CurrentFlow	Converter DC current, also called Id. It is converter's state variable, result from power flow.
idleLoss	0..1	ActivePower	Active power loss in pole at no power transfer. It is the converter's configuration data used in power flow. The attribute shall be a positive value.
maxP	0..1	ActivePower	Maximum active power limit. The value is overwritten by values of VsCapabilityCurve, if present.
maxUdc	0..1	Voltage	The maximum voltage on the DC side at which the converter should

			operate. It is the converter's configuration data used in power flow. The attribute shall be a positive value.
minP	0..1	ActivePower	Minimum active power limit. The value is overwritten by values of VsCapabilityCurve, if present.
minUdc	0..1	Voltage	The minimum voltage on the DC side at which the converter should operate. It is the converter's configuration data used in power flow. The attribute shall be a positive value.
numberOfValves	0..1	Integer	Number of valves in the converter. Used in loss calculations.
p	0..1	ActivePower	Active power at the point of common coupling. Load sign convention is used, i.e. positive sign means flow out from a node. Starting value for a steady state solution in the case a simplified power flow model is used.

poleLossP	0..1	ActivePower	The active power loss at a DC Pole = idleLoss + switchingLoss*|Idc| + resitiveLoss*Idc^2. For lossless operation Pdc=Pac. For rectifier operation with losses Pdc=Pac-lossP. For inverter operation with losses Pdc=Pac+lossP. It is converter's state variable used in power flow. The attribute shall be a positive value.
q	0..1	ReactivePower	Reactive power at the point of common coupling. Load sign convention is used, i.e. positive sign means flow out from a node. Starting value for a steady state solution in the case a simplified power flow model is used.
ratedUdc	0..1	Voltage	Rated converter DC voltage, also called UdN. The attribute shall be a positive value. It is the converter's configuration data used in power flow. For instance a bipolar DC link

			with value 200 kV has a 400kV difference between the dc lines.
resistiveLoss	0..1	Resistance	It is the converter's configuration data used in power flow. Refer to poleLossP. The attribute shall be a positive value.
switchingLoss	0..1	ActivePowerPerCurrentFlow	Switching losses, relative to the base apparent power 'baseS'. Refer to poleLossP. The attribute shall be a positive value.
targetPpcc	0..1	ActivePower	Real power injection target in AC grid, at point of common coupling. Load sign convention is used, i.e. positive sign means flow out from a node.
targetUdc	0..1	Voltage	Target value for DC voltage magnitude. The attribute shall be a positive value.
uc	0..1	Voltage	Line-to-line converter voltage, the voltage at the AC side of the valve. It is converter's state variable, result from power flow. The attribute shall be a positive value.

udc	0..1	Voltage	Converter voltage at the DC side, also called Ud. It is converter's state variable, result from power flow. The attribute shall be a positive value.
valveU0	0..1	Voltage	Valve threshold voltage, also called Uvalve. Forward voltage drop when the valve is conducting. Used in loss calculations, i.e. the switchLoss depend on numberOfValves*valveU0.

Inherited Members

BaseVoltage	0..1	BaseVoltage	see ConductingEquipment
inService	1..1	Boolean	see Equipment
EquipmentContainer	1..1	EquipmentContainer	see Equipment
mRID	1..1	String	see PowerSystemResource
name	1..1	String	see PowerSystemResource

mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

ACDCTerminal Core

An electrical connection point (AC or DC) to a piece of conducting equipment. Terminals are connected at physical connection points called connectivity nodes.

Native Members

sequenceNumber	1..1	Integer	The orientation of the terminal connections for a multiple terminal conducting equipment. The sequence numbering starts with 1 and additional terminals should follow in increasing order. The first terminal is the "starting point" for a two terminal branch.

Inherited Members

mRID	0..1	String	see IdentifiedObject

name	0..1	String	see IdentifiedObject

AsynchronousMachineDynamics

AsynchronousMachineDynamics

Asynchronous machine whose behaviour is described by reference to a standard model expressed in either time constant reactance form or equivalent circuit form or by definition of a user-defined model.

Parameter details:

1. Asynchronous machine parameters such as *Xl, Xs,* etc. are actually used as inductances in the model, but are commonly referred to as reactances since, at nominal frequency, the PU values are the same. However, some references use the symbol *L* instead of *X*.

Native Members

AsynchronousMachine	0..1	AsynchronousMachine	Asynchronous machine to which this asynchronous machine dynamics model applies.

Inherited Members

damping	1..1	Float	see RotatingMachineDynamics

inertia	1..1	Seconds	see RotatingMachineDynamics
statorLeakageReactance	1..1	PU	see RotatingMachineDynamics
statorResistance	1..1	PU	see RotatingMachineDynamics
mRID	1..1	String	see DynamicsFunctionBlock
enabled	1..1	Boolean	see DynamicsFunctionBlock
name	1..1	String	see DynamicsFunctionBlock
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Breaker

Wires

A mechanical switching device capable of making, carrying, and breaking currents under normal circuit conditions and also making, carrying for a specified time, and breaking currents under specified abnormal circuit conditions e.g. those of short circuit.

Inherited Members

BaseVoltage	0..1	BaseVoltage	see ConductingEquipment
inService	1..1	Boolean	see Equipment
EquipmentContainer	1..1	EquipmentContainer	see Equipment
mRID	1..1	String	see PowerSystemResource
name	1..1	String	see PowerSystemResource
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

ConductingEquipment Core

The parts of the AC power system that are designed to carry current or that are conductively connected through terminals.

Native Members

BaseVoltage	0..1	BaseVoltage	Base voltage of this conducting equipment. Use only when there is no voltage level container used and only one base voltage applies. For example, not used for transformers.

Inherited Members

inService	1..1	Boolean	see Equipment
EquipmentContainer	1..1	EquipmentContainer	see Equipment
mRID	1..1	String	see PowerSystemResource
name	1..1	String	see PowerSystemResource
mRID	0..1	String	see IdentifiedObject

name	0..1	String	see IdentifiedObject

Conductor

Wires

Combination of conducting material with consistent electrical characteristics, building a single electrical system, used to carry current between points in the power system.

Native Members

length	1..1	Length	Segment length for calculating line segment capabilities.

Inherited Members

BaseVoltage	0..1	BaseVoltage	see ConductingEquipment
inService	1..1	Boolean	see Equipment
EquipmentContainer	1..1	EquipmentContainer	see Equipment
mRID	1..1	String	see PowerSystemResource

name	1..1	String	see PowerSystemResource
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

ConnectedFacility

Emtiop

A collection of components that comprise a facility connected to the grid, such as a generating plant or large load.

Native Members

ACPointOfCommonCoupling	0..1	ACPointOfCommonCoupling	The connection point for this facility. It should be associated with a ConnectivityNode within the facility network model, where it may be connected to an external network model.
Equipments	0..*	Equipment	The Equipments associated with this ConnectedFacility.

Inherited Members

mRID	1..1	String	see PowerSystemResource
name	1..1	String	see PowerSystemResource
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

ConnectivityNodeContainer Core

A base class for all objects that may contain connectivity nodes or topological nodes.

Inherited Members

mRID	1..1	String	see PowerSystemResource
name	1..1	String	see PowerSystemResource
mRID	0..1	String	see IdentifiedObject

name	0..1	String	see IdentifiedObject

Curve

Core

A multi-purpose curve or functional relationship between an independent variable (X-axis) and dependent (Y-axis) variables.

Native Members

mRID	1..1	String	Master resource identifier issued by a model authority. The mRID is unique within an exchange context. Global uniqueness is easily achieved by using a UUID, as specified in IETF RFC 4122, for the mRID. The use of UUID is strongly recommended. For CIMXML data files in RDF syntax conforming to IEC 61970-552, the mRID is mapped to rdf:ID or rdf:about attributes that identify CIM object elements.
curveStyle	0..1	CurveStyle	The style or shape of the curve.

name	1..1	String	The name is any free human readable and possibly non unique text naming the object.
xMultiplier	0..1	UnitMultiplier	Multiplier for X-axis.
xUnit	0..1	UnitSymbol	The X-axis units of measure.
y1Multiplier	0..1	UnitMultiplier	Multiplier for Y1-axis.
y1Unit	0..1	UnitSymbol	The Y1-axis units of measure.

Inherited Members

mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

DCBaseTerminal DC

An electrical connection point at a piece of DC conducting equipment. DC terminals are connected at one physical DC node that may have multiple DC terminals connected. A DC node is similar to an AC connectivity node. The model requires that DC connections are distinct from AC connections.

Native Members

DCNode	1..1	DCNode	The DC connectivity node to which this DC base terminal connects with zero impedance.

Inherited Members

sequenceNumber	1..1	Integer	see ACDCTerminal
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

DCConductingEquipment DC

The parts of the DC power system that are designed to carry current or that are conductively connected through DC terminals.

Native Members

ratedCurrent	0..1	CurrentFlow	The maximum continuous current carrying capacity in amps governed by the device material and construction. The attribute shall be a positive value.
ratedUdc	0..1	Voltage	Rated DC device voltage. The attribute shall be a positive value. It is configuration data used in power flow.

Inherited Members

inService	1..1	Boolean	see Equipment
EquipmentContainer	1..1	EquipmentContainer	see Equipment
mRID	1..1	String	see PowerSystemResource
name	1..1	String	see PowerSystemResource

mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

DCSwitch

DC

A switch within the DC system.

Native Members

locked	0..1	Boolean	If true, the switch is locked. The resulting switch state is a combination of locked and DCSwitch.open attributes as follows: • locked=true and DCSwitch.open=true. The resulting state is open and locked; • locked=false and DCSwitch.open=true. The resulting state is open;

			• locked=false and DCSwitch.open=false. The resulting state is closed.
normalOpen	0..1	Boolean	The attribute is used in cases when no Measurement for the status value is present. If the DCSwitch has a status measurement the Discrete.normalValue is expected to match with the DCSwitch.normalOpen.
open	0..1	Boolean	The attribute tells if the switch is considered open when used as input to topology processing.
retained	0..1	Boolean	Branch is retained in the topological solution. The flow through retained switches will normally be calculated in power flow.

Inherited Members

ratedCurrent	0..1	CurrentFlow	see DCConductingEquipment
ratedUdc	0..1	Voltage	see DCConductingEquipment

inService	1..1	Boolean	see Equipment
EquipmentContainer	1..1	EquipmentContainer	see Equipment
mRID	1..1	String	see PowerSystemResource
name	1..1	String	see PowerSystemResource
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

DetailedModelDescriptor

DetailedModelDescription

Describes different components of a detailed model.

Native Members

mRID	0..1	String	Master resource identifier issued by a model authority. The mRID is unique within an exchange context. Global uniqueness is easily

			achieved by using a UUID, as specified in IETF RFC 4122, for the mRID. The use of UUID is strongly recommended. For CIMXML data files in RDF syntax conforming to IEC 61970-552, the mRID is mapped to rdf:ID or rdf:about attributes that identify CIM object elements.
DetailedModelTypeDynamics	0..1	DetailedModelTypeDynamics	The detailed model type dynamics that has detailed model descriptor.

Inherited Members

mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

DetailedModelTypeDynamics DetailedModelDescription

The main class that packages all related to this type of a detailed model. This includes all parameters, functions, signals, etc.

DiagramObject

An object that defines one or more points in a given space. This object can be associated with anything that specializes IdentifiedObject. For single line diagrams such objects typically include such items as analog values, breakers, disconnectors, power transformers, and transmission lines.

Native Members

mRID	1..1	String	Master resource identifier issued by a model authority. The mRID is unique within an exchange context. Global uniqueness is easily achieved by using a UUID, as specified in IETF RFC 4122, for the mRID. The use of UUID is strongly recommended. For CIMXML data files in RDF syntax conforming to IEC 61970-552, the mRID is mapped to rdf:ID or rdf:about attributes that identify CIM object elements.
drawingOrder	1..1	Integer	The drawing order of this element. The higher the number, the later the element is drawn in sequence. This is used to ensure that elements that overlap are rendered in the correct order.

isPolygon	1..1	Boolean	Defines whether or not the diagram objects points define the boundaries of a polygon or the routing of a polyline. If this value is true then a receiving application should consider the first and last points to be connected.
name	1..1	String	The name is any free human readable and possibly non unique text naming the object.
IdentifiedObject	1..1	IdentifiedObject	The domain object to which this diagram object is associated.

Inherited Members

mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

DynamicsFunctionBlock

StandardModels

Abstract parent class for all Dynamics function blocks.

Native Members

mRID	1..1	String	Master resource identifier issued by a model authority. The mRID is unique within an exchange context. Global uniqueness is easily achieved by using a UUID, as specified in IETF RFC 4122, for the mRID. The use of UUID is strongly recommended. For CIMXML data files in RDF syntax conforming to IEC 61970-552, the mRID is mapped to rdf:ID or rdf:about attributes that identify CIM object elements.
enabled	1..1	Boolean	Function block used indicator. true = use of function block is enabled false = use of function block is disabled.
name	1..1	String	The name is any free human readable and possibly non unique text naming the object.

Inherited Members

mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

EnergyConnection

Wires

A connection of energy generation or consumption on the power system model.

Inherited Members

BaseVoltage	0..1	BaseVoltage	see ConductingEquipment
inService	1..1	Boolean	see Equipment
EquipmentContainer	1..1	EquipmentContainer	see Equipment
mRID	1..1	String	see PowerSystemResource
name	1..1	String	see PowerSystemResource

mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Equipment

Core

The parts of a power system that are physical devices, electronic or mechanical.

Native Members

inService	1..1	Boolean	Specifies the availability of the equipment. True means the equipment is available for topology processing, which determines if the equipment is energized or not. False means that the equipment is treated by network applications as if it is not in the model.
EquipmentContainer	1..1	EquipmentContainer	Container of this equipment.

Inherited Members

mRID	1..1	String	see PowerSystemResource
name	1..1	String	see PowerSystemResource
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

GeneratingUnit Production

A single or set of synchronous machines for converting mechanical power into alternating-current power. For example, individual machines within a set may be defined for scheduling purposes while a single control signal is derived for the set. In this case there would be a GeneratingUnit for each member of the set and an additional GeneratingUnit corresponding to the set.

Native Members

maxOperatingP	1..1	ActivePower	This is the maximum operating active power limit the dispatcher can enter for this unit.

minOperatingP	1..1	ActivePower	This is the minimum operating active power limit the dispatcher can enter for this unit.

Inherited Members

inService	1..1	Boolean	see Equipment
EquipmentContainer	1..1	EquipmentContainer	see Equipment
mRID	1..1	String	see PowerSystemResource
name	1..1	String	see PowerSystemResource
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

IEEECigreAPISignal Emtiop

The parent class for CIGRE TB 958 input and output signals. Use the ACDCTerminal association for network flows, the DCNode association for DC bus quantities, the ConnectivityNode association for AC bus quantities, or no association for references levels or other signals not connected to the electric power networks.

Native Members

apiName	0..1	String	The name of this signal, as returned by the CIGRE TB 958 API. The inherited IdentifiedObject.name attribute might not necessarily match this name.
apiParameterKind	0..1	IEEECigreAPIParameterKind	Establishes the signal value size, in bytes, expected in the CIGRE TB 958 API.
apiSequenceNumber	0..1	Integer	The signal's expected zero-based sequence number in the CIGRE TB 958 API array for input and output signals.
apiWidth	0..1	Integer	Signal array dimension from the CIGRE TB 958 API, defaults to 1.
multiplier	0..1	UnitMultiplier	Multiplier for the units of this signal, in CIM. May require interpretation of information returned from the CIGRE TB 958 API.

phase	0..1	SinglePhaseKind	The signal's phase, as applicable, for multiphase signal connections.
unit	0..1	UnitSymbol	Signal units, if applicable, in CIM. May require interpretation of information returned from the CIGRE TB 958 API.
ConnectivityNode	0..1	ConnectivityNode	Use for a bus voltage or other bus quantity signal. Mutually exclusive with association to DCNode or ACDCTerminal.
DCNode	0..1	DCNode	Use for a DC voltage or other quantity related to DC buses. Mutually exclusive with assocation to ConnectivityNode or ACDCTerminal.
IEEECigreAPISignalInfo	0..1	IEEECigreAPISignalInfo	Expanded set of attributes available from the CIGRE TB 958 API.

Inherited Members

ACDCTerminal	0..1	ACDCTerminal	see SignalDescriptor

DynamicsFunctionBlock	0..1	DynamicsFunctionBlock	see SignalDescriptor
mRID	0..1	String	see DetailedModelDescriptor
DetailedModelTypeDynamics	0..1	DetailedModelTypeDynamics	see DetailedModelDescriptor
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

IdentifiedObject Core

This is a class that provides common identification for all classes needing identification and naming attributes.

Native Members

mRID	0..1	String	Master resource identifier issued by a model authority. The mRID is unique within an exchange context. Global uniqueness is easily achieved by using a UUID, as specified in IETF RFC 4122, for the mRID. The

			use of UUID is strongly recommended. For CIMXML data files in RDF syntax conforming to IEC 61970-552, the mRID is mapped to rdf:ID or rdf:about attributes that identify CIM object elements.
name	0..1	String	The name is any free human readable and possibly non unique text naming the object.

MutualCoupling

Wires

This class represents the zero sequence line mutual coupling.

Native Members

mRID	1..1	String	Master resource identifier issued by a model authority. The mRID is unique within an exchange context. Global uniqueness is easily achieved by using a UUID, as specified in IETF RFC 4122, for the mRID. The use of UUID is strongly recommended.

			For CIMXML data files in RDF syntax conforming to IEC 61970-552, the mRID is mapped to rdf:ID or rdf:about attributes that identify CIM object elements.
b0ch	0..1	Susceptance	Zero sequence mutual coupling shunt (charging) susceptance, uniformly distributed, of the entire line section.
distance11	0..1	Length	Distance to the start of the coupled region from the first line's terminal having sequence number equal to 1.
distance12	0..1	Length	Distance to the end of the coupled region from the first line's terminal with sequence number equal to 1.
distance21	0..1	Length	Distance to the start of coupled region from the second line's terminal with sequence number equal to 1.
distance22	0..1	Length	Distance to the end of coupled region from the second line's

			terminal with sequence number equal to 1.
g0ch	0..1	Conductance	Zero sequence mutual coupling shunt (charging) conductance, uniformly distributed, of the entire line section.
name	1..1	String	The name is any free human readable and possibly non unique text naming the object.
r0	0..1	Resistance	Zero sequence branch-to-branch mutual impedance coupling, resistance.
x0	0..1	Reactance	Zero sequence branch-to-branch mutual impedance coupling, reactance.

Inherited Members

mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

OperationalLimit

A value and normal value associated with a specific kind of limit.

The sub class value and normalValue attributes vary inversely to the associated OperationalLimitType.acceptableDuration (acceptableDuration for short).

If a particular piece of equipment has multiple operational limits of the same kind (apparent power, current, etc.), the limit with the greatest acceptableDuration shall have the smallest limit value and the limit with the smallest acceptableDuration shall have the largest limit value. Note: A large current can only be allowed to flow through a piece of equipment for a short duration without causing damage, but a lesser current can be allowed to flow for a longer duration.

Native Members

mRID	1..1	String	Master resource identifier issued by a model authority. The mRID is unique within an exchange context. Global uniqueness is easily achieved by using a UUID, as specified in IETF RFC 4122, for the mRID. The use of UUID is strongly recommended. For CIMXML data files in RDF syntax conforming to IEC 61970-552, the mRID is mapped to rdf:ID or rdf:about attributes that identify CIM object elements.

name	1..1	String	The name is any free human readable and possibly non unique text naming the object.
OperationalLimitSet	0..1	OperationalLimitSet	The limit set to which the limit values belong.
OperationalLimitType	0..1	OperationalLimitType	The limit type associated with this limit.

Inherited Members

mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

PSRType Core

Classifying instances of the same class, e.g. overhead and underground ACLineSegments. This classification mechanism is intended to provide flexibility outside the scope of this document, i.e. provide customisation that is non standard.

Inherited Members

mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

PhaseTapChanger

Wires

A transformer phase shifting tap model that controls the phase angle difference across the power transformer and potentially the active power flow through the power transformer. This phase tap model may also impact the voltage magnitude.

Native Members

TransformerEnd	0..1	TransformerEnd	Transformer end to which this phase tap changer belongs.

Inherited Members

highStep	0..1	Integer	see TapChanger
lowStep	0..1	Integer	see TapChanger

neutralStep	0..1	Integer	see TapChanger
neutralU	0..1	Voltage	see TapChanger
normalStep	0..1	Integer	see TapChanger
step	1..1	Float	see TapChanger
mRID	1..1	String	see PowerSystemResource
name	1..1	String	see PowerSystemResource
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

PointOfCommonCoupling Core

Point of Common Coupling (PCC) refers to the location where multiple electrical sources or loads are electrically connected and provide a reference point where the voltages and currents from different parts of the system are considered to be common. The PCC is used to support system analysis, control, and monitoring, as it provides a reference for understanding the interactions and power flow between various components within the system. It is also relevant to define the requirement and responsibility between different actors in operating a power system.

Native Members

mRID	1..1	String	Master resource identifier issued by a model authority. The mRID is unique within an exchange context. Global uniqueness is easily achieved by using a UUID, as specified in IETF RFC 4122, for the mRID. The use of UUID is strongly recommended. For CIMXML data files in RDF syntax conforming to IEC 61970-552, the mRID is mapped to rdf:ID or rdf:about attributes that identify CIM object elements.
name	1..1	String	The name is any free human readable and possibly non unique text naming the object.

Inherited Members

mRID	0..1	String	see IdentifiedObject

name	0..1	String	see IdentifiedObject

PowerElectronicsUnit

Production

A generating unit or battery or aggregation that connects to the AC network using power electronics rather than rotating machines.

Native Members

maxP	1..1	ActivePower	Maximum active power limit. This is the maximum (nameplate) limit for the unit.
minP	1..1	ActivePower	Minimum active power limit. This is the minimum (nameplate) limit for the unit.
PowerElectronicsConnection	1..1	PowerElectronicsConnection	A power electronics unit has a connection to the AC network.

Inherited Members

inService	1..1	Boolean	

			see Equipment
EquipmentContainer	1..1	EquipmentContainer	see Equipment
mRID	1..1	String	see PowerSystemResource
name	1..1	String	see PowerSystemResource
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

PowerSystemResource

Core

A power system resource (PSR) can be an item of equipment such as a switch, an equipment container containing many individual items of equipment such as a substation, or an organisational entity such as sub-control area. Power system resources can have measurements associated.

Native Members

mRID	1..1	String	Master resource identifier issued by a model authority. The mRID is

			unique within an exchange context. Global uniqueness is easily achieved by using a UUID, as specified in IETF RFC 4122, for the mRID. The use of UUID is strongly recommended. For CIMXML data files in RDF syntax conforming to IEC 61970-552, the mRID is mapped to rdf:ID or rdf:about attributes that identify CIM object elements.
name	1..1	String	The name is any free human readable and possibly non unique text naming the object.

Inherited Members

mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

ProtectedSwitch

Wires

A ProtectedSwitch is a switching device that can be operated by ProtectionEquipment.

Inherited Members

BaseVoltage	0..1	BaseVoltage	see ConductingEquipment
inService	1..1	Boolean	see Equipment
EquipmentContainer	1..1	EquipmentContainer	see Equipment
mRID	1..1	String	see PowerSystemResource
name	1..1	String	see PowerSystemResource
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

RegulatingCondEq Wires

A type of conducting equipment that can regulate a quantity (i.e. voltage or flow) at a specific point in the network.

Inherited Members

BaseVoltage	0..1	BaseVoltage	see ConductingEquipment
inService	1..1	Boolean	see Equipment
EquipmentContainer	1..1	EquipmentContainer	see Equipment
mRID	1..1	String	see PowerSystemResource
name	1..1	String	see PowerSystemResource
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

RotatingMachine

Wires

A rotating machine which may be used as a generator or motor.

Native Members

p	1..1	ActivePower	Active power injection. Load sign convention is used, i.e. positive sign means flow out from a node. Starting value for a steady state solution.
q	1..1	ReactivePower	Reactive power injection. Load sign convention is used, i.e. positive sign means flow out from a node. Starting value for a steady state solution.
ratedS	1..1	ApparentPower	Nameplate apparent power rating for the unit. The attribute shall have a positive value.
ratedU	1..1	Voltage	Rated voltage (nameplate data, Ur in IEC 60909-0). It is primarily used for short circuit data exchange according to IEC 60909. The attribute shall be a positive value.

GeneratingUnit	1..1	GeneratingUnit	A synchronous machine may operate as a generator and as such becomes a member of a generating unit.

Inherited Members

BaseVoltage	0..1	BaseVoltage	see ConductingEquipment
inService	1..1	Boolean	see Equipment
EquipmentContainer	1..1	EquipmentContainer	see Equipment
mRID	1..1	String	see PowerSystemResource
name	1..1	String	see PowerSystemResource
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

RotatingMachineDynamics

StandardModels

Abstract parent class for all synchronous and asynchronous machine standard models.

Native Members

damping	1..1	Float	Damping torque coefficient (D) (>= 0) in pu torque/pu speed deviation (differential type). A proportionality constant that, when multiplied by the angular velocity of the rotor poles with respect to the magnetic field (frequency), results in the damping torque. This value is often zero when the sources of damping torques (generator damper windings, load damping effects, etc.) are modelled in detail. Typical value = 0.
inertia	1..1	Seconds	Inertia constant of generator or motor and mechanical load (*H*) (> 0). This is the specification for the stored energy in the rotating mass when operating at rated speed. For a generator, this includes the generator plus all other elements (turbine, exciter) on the same shaft and has units of MW x s. For a motor, it includes the motor plus its mechanical load. Conventional units

			are PU on the generator MVA base, usually expressed as MW x s / MVA or just s. This value is used in the accelerating power reference frame for operator training simulator solutions. Typical value = 3.
statorLeakageReactance	1..1	PU	Stator leakage reactance (*Xl*) (>= 0). Typical value = 0,15.
statorResistance	1..1	PU	Stator (armature) resistance (*Rs*) (>= 0). Typical value = 0,005.

Inherited Members

mRID	1..1	String	see DynamicsFunctionBlock
enabled	1..1	Boolean	see DynamicsFunctionBlock
name	1..1	String	see DynamicsFunctionBlock
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

ShuntCompensator

Wires

A shunt capacitor or reactor or switchable bank of shunt capacitors or reactors. A section of a shunt compensator is an individual capacitor or reactor. A negative value for bPerSection indicates that the compensator is a reactor. ShuntCompensator is a single terminal device. Ground is implied.

Native Members

grounded	1..1	Boolean	Required for Yn and I connections (as represented by ShuntCompensator.phaseConnection). True if the neutral is solidly grounded.
maximumSections	1..1	Integer	The maximum number of sections that may be switched in.
nomU	1..1	Voltage	The voltage at which the nominal reactive power may be calculated. This should normally be within 10% of the voltage at which the capacitor is connected to the network.
phaseConnection	0..1	PhaseShuntConnectionKind	The type of phase connection, such as wye or delta.
sections	1..1	Float	Shunt compensator sections in use. Starting value for steady state solution. The attribute shall

			be a positive value or zero. Non integer values are allowed to support continuous variables. The reasons for continuous value are to support study cases where no discrete shunt compensators has yet been designed, a solutions where a narrow voltage band force the sections to oscillate or accommodate for a continuous solution as input. For LinearShuntConpensator the value shall be between zero and ShuntCompensator.maximumSections. At value zero the shunt compensator conductance and admittance is zero. Linear interpolation of conductance and admittance between the previous and next integer section is applied in case of non-integer values. For NonlinearShuntCompensator(-s) shall only be set to one of the NonlinearShuntCompenstorPoint.sectionNumber. There is no interpolation between NonlinearShuntCompenstorPoint(-s).

Inherited Members

BaseVoltage	0..1	BaseVoltage	see ConductingEquipment
inService	1..1	Boolean	see Equipment

EquipmentContainer	1..1	EquipmentContainer	see Equipment
mRID	1..1	String	see PowerSystemResource
name	1..1	String	see PowerSystemResource
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

SignalDescriptor DetailedModelDescription

Describes the signals both internal signals that connect different functions or external signals.

Native Members

ACDCTerminal	0..1	ACDCTerminal	The terminal for this signal descriptor.
DynamicsFunctionBlock	0..1	DynamicsFunctionBlock	The dynamics function block to which this signal belongs to.

Inherited Members

mRID	0..1	String	see DetailedModelDescriptor
DetailedModelTypeDynamics	0..1	DetailedModelTypeDynamics	see DetailedModelDescriptor
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

StateVariable

StateVariables

An abstract class for state variables.

Switch

Wires

A generic device designed to close, or open, or both, one or more electric circuits. All switches are two terminal devices including grounding switches. The ACDCTerminal.connected at the two sides of the switch shall not be considered for assessing switch connectivity, i.e. only Switch.open, .normalOpen and .locked are relevant.

Inherited Members

BaseVoltage	0..1	BaseVoltage	see ConductingEquipment
inService	1..1	Boolean	see Equipment
EquipmentContainer	1..1	EquipmentContainer	see Equipment
mRID	1..1	String	see PowerSystemResource
name	1..1	String	see PowerSystemResource
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

SynchronousMachineDetailed SynchronousMachineDynamics

All synchronous machine detailed types use a subset of the same data parameters and input/output variables.

The several variations differ in the following ways:

- the number of equivalent windings that are included;

- the way in which saturation is incorporated into the model;

- whether or not â€œsubtransient saliencyâ€� (*X''q* not = *X''d*) is represented.

It is not necessary for each simulation tool to have separate models for each of the model types. The same model can often be used for several types by alternative logic within the model. Also, differences in saturation representation might not result in significant model performance differences so model substitutions are often acceptable.

Native Members

efdBaseRatio	1..1	Float	Ratio (exciter voltage/generator voltage) of *Efd* bases of exciter and generator models (> 0). Typical value = 1.
ifdBaseType	1..1	IfdBaseKind	Excitation base system mode. It should be equal to the value of *WLMDV* given by the user. *WLMDV* is the PU ratio between the field voltage and the excitation current: *Efd* = *WLMDV* x *Ifd*. Typical value = ifag.
saturationFactor	1..1	Float	Saturation factor at rated terminal voltage (*S1*) (>= 0). Defined by defined by *S*(*E1*) in the SynchronousMachineSaturationParameters diagram. Typical value = 0,02.
saturationFactor120	1..1	Float	Saturation factor at 120 % of rated terminal voltage (*S12*) (>=

			RotatingMachineDynamics.saturationFactor). Defined by *S*(*E2*) in the SynchronousMachineSaturationParameters diagram. Typical value = 0,12.
saturationFactor120QAxis	1..1	Float	Quadrature-axis saturation factor at 120% of rated terminal voltage (*S12q*) (>= SynchonousMachineDetailed.saturationFactorQAxis). Typical value = 0,12.
saturationFactorQAxis	1..1	Float	Quadrature-axis saturation factor at rated terminal voltage (*S1q*) (>= 0). Typical value = 0,02.

Inherited Members

SynchronousMachine	0..1	SynchronousMachine	see SynchronousMachineDynamics
damping	1..1	Float	see RotatingMachineDynamics
inertia	1..1	Seconds	see RotatingMachineDynamics
statorLeakageReactance	1..1	PU	see RotatingMachineDynamics

statorResistance	1..1	PU	see RotatingMachineDynamics
mRID	1..1	String	see DynamicsFunctionBlock
enabled	1..1	Boolean	see DynamicsFunctionBlock
name	1..1	String	see DynamicsFunctionBlock
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

SynchronousMachineDynamics SynchronousMachineDynamics

Synchronous machine whose behaviour is described by reference to a standard model expressed in one of the following forms:

- simplified (or classical), where a group of generators or motors is not modelled in detail;

- detailed, in equivalent circuit form;

- detailed, in time constant reactance form; or

- by definition of a user-defined model.

It is a common practice to represent small generators by a negative load rather than by a dynamic generator model when performing dynamics simulations. In this case, a SynchronousMachine in the static model is not represented by anything in the dynamics model, instead it is treated as an ordinary load.

Parameter details:

1. Synchronous machine parameters such as *Xl, Xd, Xp* etc. are actually used as inductances in the models, but are commonly referred to as reactances since, at nominal frequency, the PU values are the same. However, some references use the symbol *L* instead of *X*.

Native Members

SynchronousMachine	0..1	SynchronousMachine	Synchronous machine to which synchronous machine dynamics model applies.

Inherited Members

damping	1..1	Float	see RotatingMachineDynamics
inertia	1..1	Seconds	see RotatingMachineDynamics
statorLeakageReactance	1..1	PU	see RotatingMachineDynamics
statorResistance	1..1	PU	

			see RotatingMachineDynamics
mRID	1..1	String	see DynamicsFunctionBlock
enabled	1..1	Boolean	see DynamicsFunctionBlock
name	1..1	String	see DynamicsFunctionBlock
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

TapChanger

Wires

Mechanism for changing transformer winding tap positions.

Native Members

highStep	0..1	Integer	Highest possible tap step position, advance from neutral.

			The attribute shall be greater than lowStep.
lowStep	0..1	Integer	Lowest possible tap step position, retard from neutral.
neutralStep	0..1	Integer	The neutral tap step position for this winding. The attribute shall be equal to or greater than lowStep and equal or less than highStep. It is the step position where the voltage is neutralU when the other terminals of the transformer are at the ratedU. If there are other tap changers on the transformer those taps are kept constant at their neutralStep.
neutralU	0..1	Voltage	Voltage at which the winding operates at the neutral tap setting. It is the voltage at the terminal of the PowerTransformerEnd associated with the tap changer when all tap changers on the transformer are at their neutralStep position. Normally neutralU of the tap changer is the same as ratedU of the PowerTransformerEnd, but it can

			differ in special cases such as when the tapping mechanism is separate from the winding more common on lower voltage transformers. This attribute is not relevant for PhaseTapChangerAsymmetrical, PhaseTapChangerSymmetrical and PhaseTapChangerLinear.
normalStep	0..1	Integer	The tap step position used in "normal" network operation for this winding. For a "Fixed" tap changer indicates the current physical tap setting. The attribute shall be equal to or greater than lowStep and equal to or less than highStep.
step	1..1	Float	Tap changer position. Starting step for a steady state solution. Non integer values are allowed to support continuous tap variables. The reasons for continuous value are to support study cases where no discrete tap changer has yet been designed, a solution where a narrow voltage band forces the tap step to oscillate

or to accommodate for a continuous solution as input.

The attribute shall be equal to or greater than lowStep and equal to or less than highStep.

Inherited Members

mRID	1..1	String	see PowerSystemResource
name	1..1	String	see PowerSystemResource
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

TransformerEnd Wires

A conducting connection point of a power transformer. It corresponds to a physical transformer winding terminal. In earlier CIM versions, the TransformerWinding class served a similar purpose, but this class is more flexible because it associates to terminal but is not a specialization of ConductingEquipment.

Native Members

mRID	1..1	String	Master resource identifier issued by a model authority. The mRID is unique within an exchange context. Global uniqueness is easily achieved by using a UUID, as specified in IETF RFC 4122, for the mRID. The use of UUID is strongly recommended. For CIMXML data files in RDF syntax conforming to IEC 61970-552, the mRID is mapped to rdf:ID or rdf:about attributes that identify CIM object elements.
endNumber	1..1	Integer	Number for this transformer end, corresponding to the end's order in the power transformer vector group or phase angle clock number. Highest voltage winding should be 1. Each end within a power transformer should have a unique subsequent end number. Note the transformer end number need not match the terminal sequence number.
grounded	1..1	Boolean	Used only for Yn and Zn connections indicated by PowerTransformerEnd.connectionKind. If true, the neutral is grounded and

			attributes TransformerEnd.rground and TransformerEnd.xground are required. If false, the attributes TransformerEnd.rground and TransformerEnd.xground are not considered.
name	1..1	String	The name is any free human readable and possibly non unique text naming the object.
rground	1..1	Resistance	Resistance part of neutral impedance. Zero indicates solidly grounded or grounded through a reactor.
xground	1..1	Reactance	Reactance part of neutral impedance. Zero indicates solidly grounded or grounded through a reactor.
BaseVoltage	1..1	BaseVoltage	Base voltage of the transformer end. This is essential for PU calculation.
Terminal	1..1	Terminal	Terminal of the power transformer to which this transformer end belongs.

Inherited Members

mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Enumerations

AsynchronousMachineKind Wires

Kind of Asynchronous Machine.

generator	The Asynchronous Machine is a generator.
motor	The Asynchronous Machine is a motor.

BatteryStateKind Production

The state of the battery unit.

charging	
discharging	
empty	
full	
waiting	

CsOperatingModeKind DC

Operating mode for DC line operating as Current Source Converter.

inverter	Operating as inverter, which is the power receiving end.
rectifier	Operating as rectifier, which is the power sending end.

CsPpccControlKind DC

Active power control modes for DC line operating as Current Source Converter.

activePower	Control is active power control at AC side, at point of common coupling. Target is provided by ACDCConverter.targetPpcc.
dcCurrent	Control is DC current with target value provided by CsConverter.targetIdc.
dcVoltage	Control is DC voltage with target value provided by ACDCConverter.targetUdc.

CurveStyle

Core

Style or shape of curve.

constantYValue	
straightLineYValues	

DCPolarityKind

DC

Polarity for DC circuits.

middle	Middle pole. The converter terminal is the midpoint in a bipolar or symmetric monopole configuration. The midpoint can be grounded and/or have a metallic return.
negative	Negative pole. The converter terminal is intended to operate at a negative voltage relative the midpoint or positive terminal.
positive	Positive pole. The converter terminal is intended to operate at a positive voltage relative the midpoint or negative terminal.

DCSourceKind Emtiop

battery	Represent the DCEnergySource with a nonlinear battery model, which should respond to state-of-charge (SoC) controls.
load	For electronic loads. Passive loads can be represented in DCShunt.
photoVoltaic	Represent the DCEnergySource with a nonlinear PV panel model, which should respond to maximum power point tracking (MPPT) control

DCTerminalPolarityKind DC

Polarity for DC terminal. Used in DC system configurations that have explicit polarity of the terminals, e.g., voltage source converter (VSC) technology.

negative	Negative terminal.
positive	Positive terminal.

IEEECigreAPIInputKind Emtiop

acCurrent	AC current from inverter into the AC filter. Requires the phase attribute. Typically in Amperes, but the CIGRE TB 958 API should be used to verify units.
acCurrentGrid	AC current from AC filter into the grid. Requires the phase attribute. Typically in Amperes, but the CIGRE TB 958 API should be used to verify units.
acVoltage	AC voltage at the filter-to-grid connection point. Requires the phase attribute. Typically in Volts, but the CIGRE TB 958 API should be used to verify units.
activePowerReference	Active power control reference. Typically in per-unit but the CIGRE TB 958 API should be used to verify units.
apiDefined	Another kind of input or control signal not enumerated in CIM. Use the CIGRE TB 958 API for more information.

dcCurrent	DC current into the inverter stage, if DC bus modeling applies. Typically in Amperes, but the DLL API should be used to verify units.
dcMPPTVoltage	DC voltage command from the maximum power point tracking system, if DC bus modeling applies. Typically in Volts, but the DLL API should be used to verify units.
dcVoltage	DC voltage at the inverter stage, if DC bus modeling applies. Typically in Volts, but the CIGRE TB 958 API should be used to verify units.
reactivePowerReference	Reactive power control reference. Typically in per-unit but the CIGRE TB 958 API should be used to verify units.
voltageReference	Voltage control reference. Typically in per-unit and positive sequence, but the CIGRE TB 958 API should be used to verify units.

IEEECigreAPIModeKind Emtiop

SupportsBoth	The CIGRE TB 958 model runs in either EMT or RMS simulations.
SupportsEMT	The CIGRE TB 958 model runs in EMT but not RMS simulations.

`SupportsNone`	This CIGRE TB 958 model is unusable.
`SupportsRMS`	The CIGRE TB 958 model runs in RMS simulation, e.g., power flow, short-circuit, positive sequence dynamics, transient stability. It does not run in EMT simulation.

IEEECigreAPIOutputKind Emtiop

`activePower`	Active power from internal calculation; a convenience output.
`apiDefined`	Typically a convenience output for plotting and analysis. Use CIGRE TB 958 API for more information.
`modulationIndex`	Modulation index for PWM switching in a detailed VSC model. May be scaled by Vdc/2 in an average model. Requires the phase attribute. If these outputs are not provided, then vscVoltage outputs shall be provided.
`pllFrequency`	Frequency estimated from the model's phase locked loop or similar algorithm.
`reactivePower`	Reactive power from internal calculation; a convenience output.
`rideThroughMode`	A flag indicating fault-ride-through mode is active, based on logic internal to the model.

`vscVoltage`	VSC source voltage for an average model. Requires the phase attribute. If these outputs are not provided then modulationIndex outputs shall be provided.

IEEECigreAPIParameterKind Emtiop

Indicates the type and size (bytes) of a parameter, as documented in IEEE_Cigre_DLLInterface_types.h from CIGRE TB 958.

`Char_Ptr`
`Char_Val`
`Int16_Val`
`Int32_Val`
`Int8_Val`
`Real32_Val`
`Real64_Val`

Uint16_Val

Uint32_Val

Uint8_Val

IfdBaseKind SynchronousMachineDynamics

Excitation base system mode.

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ifag

---

iffl

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ifnl

InputSignalKind PowerSystemStabilizerDynamics

Types of input signals. In dynamics modelling, commonly represented by the *j* parameter.

branchCurrent
busFrequency
busFrequencyDeviation
busVoltage
busVoltageDerivative
fieldCurrent
generatorAcceleratingPower
generatorElectricalPower
generatorMechanicalPower
rotorAngularFrequencyDeviation
rotorSpeed

LimitKind

Limit kinds.

alarmVoltage	Voltage alarm.
highVoltage	Referring to the rating of the equipments, a voltage too high can lead to accelerated ageing or the destruction of the equipment. This limit type may or may not have duration.
lowVoltage	A too low voltage can disturb the normal operation of some protections and transformer equipped with on-load tap changers, electronic power devices or can affect the behaviour of the auxiliaries of generation units. This limit type may or may not have duration.
operationalVoltageLimit	Operational voltage limit.
patl	The Permanent Admissible Transmission Loading (PATL) is the loading in amperes, MVA or MW that can be accepted by a network branch for an unlimited duration without any risk for the material. The OperationnalLimitType.isInfiniteDuration is set to true. There shall be only one OperationalLimitType of kind PATL per OperationalLimitSet if the PATL is

	ApparentPowerLimit, ActivePowerLimit, or CurrentLimit for a given Terminal or Equipment.
patlt	Permanent Admissible Transmission Loading Threshold (PATLT) is a value in engineering units defined for PATL and calculated using a percentage less than 100 % of the PATL type intended to alert operators of an arising condition. The percentage should be given in the name of the OperationalLimitSet. The aceptableDuration is another way to express the severity of the limit.
stability	Stability.
tatl	Temporarily Admissible Transmission Loading (TATL) which is the loading in amperes, MVA or MW that can be accepted by a branch for a certain limited duration. The TATL can be defined in different ways: • as a fixed percentage of the PATL for a given time (for example, 115% of the PATL that can be accepted during 15 minutes), • pairs of TATL type and Duration calculated for each line taking into account its particular configuration and conditions of functioning (for example, it can define a TATL acceptable during 20 minutes and another one acceptable during 10 minutes). Such a definition of TATL can depend on the initial operating conditions of the network element (sag situation of a line). The duration attribute can be used to define several TATL limit types. Hence multiple TATL limit values may exist having different durations.

tc	Tripping Current (TC) is the ultimate intensity without any delay. It is defined as the threshold the line will trip without any possible remedial actions. The tripping of the network element is ordered by protections against short circuits or by overload protections, but in any case, the activation delay of these protections is not compatible with the reaction delay of an operator (less than one minute). The duration is always zero if the OperationalLimitType.acceptableDuration is exchanged. Only one limit value exists for the TC type.
tct	Tripping Current Threshold (TCT) is a value in engineering units defined for TC and calculated using a percentage less than 100 % of the TC type intended to alert operators of an arising condition. The percentage should be given in the name of the OperationalLimitSet. The aceptableDuration is another way to express the severity of the limit.
warningVoltage	Voltage warning.

NthAmModelKind Emtiop

Application of this model.

currentCompensation	

excitationSystem

load

loadController

machine

powerSystemStabilizer

protection

renewableEnergyResource

signalPlayback

staticVarAndFACTS

turbineGovernor

NthAmModelNameKind

Emtiop

Describes the naming category for dynamic models recognized by NERC.

AUX

DGS

DYD

DYR

Other

NthAmModelStatusKind

Emtiop

allowed	NERC allows this model for interconnection-wide studies. NERC used to keep a list of allowed models; currently, it lists only prohibited models.
deprecated	NERC allows this model in interconnection-wide studies, but other models are more suitable.

prohibited	NERC prohibits use of this model in interconnection-side studies. Some legacy network examples may still use this model.

OperationalLimitDirectionKind OperationalLimits

The direction attribute describes the side of a limit that is a violation.

absoluteValue	An absoluteValue limit means that a monitored absolute value above the limit value is a violation.
high	High means that a monitored value above the limit value is a violation. If applied to a terminal flow, the positive direction is into the terminal.
low	Low means a monitored value below the limit is a violation. If applied to a terminal flow, the positive direction is into the terminal.

PhaseShuntConnectionKind Wires

The configuration of phase connections for a single terminal device such as a load or capacitor.

	Delta connection.

D	
G	Ground connection; use when explicit connection to ground needs to be expressed in combination with the phase code, such as for electrical wire/cable or for meters.
I	Independent winding, for single-phase connections.
Y	Wye connection.
Yn	Wye, with neutral brought out for grounding.

RotorKind

SynchronousMachineDynamics

Type of rotor on physical machine.

roundRotor	
salientPole	

SVCControlMode

Wires

Static VAr Compensator control mode.

reactivePower	Reactive power control.
voltage	Voltage control.

SinglePhaseKind

Wires

Enumeration of phase identifiers used to designate the specific phase of conducting equipment modelled as individual unbalanced phases.

Allows designation of specific phases for transmission and distribution equipment, circuits and loads.

A	Phase A.
B	Phase B.
C	Phase C.
N	Neutral.

s1	Secondary phase 1.
s2	Secondary phase 2.

SynchronousMachineKind

Wires

Synchronous machine type.

condenser
generator
generatorOrCondenser
generatorOrCondenserOrMotor
generatorOrMotor
motor

motorOrCondenser

SynchronousMachineModelKind SynchronousMachineDynamics

Type of synchronous machine model used in dynamic simulation applications.

subtransient

subtransientSimplified

subtransientSimplifiedDirectAxis

subtransientTypeF

subtransientTypeJ

SynchronousMachineOperatingMode Wires

Synchronous machine operating mode.

condenser

generator

motor

UnitMultiplier

Domain

The unit multipliers defined for the CIM. When applied to unit symbols, the unit symbol is treated as a derived unit. Regardless of the contents of the unit symbol text, the unit symbol shall be treated as if it were a single-character unit symbol. Unit symbols should not contain multipliers, and it should be left to the multiplier to define the multiple for an entire data type.

For example, if a unit symbol is "m2Pers" and the multiplier is "k", then the value is k(m**2/s), and the multiplier applies to the entire final value, not to any individual part of the value. This can be conceptualized by substituting a derived unit symbol for the unit type. If one imagines that the symbol "Þ" represents the derived unit "m2Pers", then applying the multiplier "k" can be conceptualized simply as "kÞ".

For example, the SI unit for mass is "kg" and not "g". If the unit symbol is defined as "kg", then the multiplier is applied to "kg" as a whole and does not replace the "k" in front of the "g". In this case, the multiplier of "m" would be used with the unit symbol of "kg" to represent one gram. As a text string, this violates the instructions in IEC 80000-1. However, because the unit symbol in CIM is treated as a derived unit instead of as an SI unit, it makes more sense to conceptualize the "kg" as if it were replaced by one of the proposed replacements for the SI mass symbol. If one imagines that the "kg" were replaced by a symbol "Þ", then it is easier to conceptualize the multiplier "m" as creating the proper unit "mÞ", and not the forbidden unit "mkg".

E
G
M
P
T
Y
Z
a
c
d
da

f
h
k
m
micro
n
none
p
y
z

UnitSymbol

Domain

The derived units defined for usage in the CIM. In some cases, the derived unit is equal to an SI unit. Whenever possible, the standard derived symbol is used instead of the formula for the derived unit. For example, the unit symbol Farad is defined as "F" instead of "CPerV". In cases where a standard symbol does not exist for a derived unit, the formula for the unit is used as the unit symbol. For example, density does not have a standard symbol and so it is represented as "kgPerm^3". With the exception of the "kg", which is an SI unit, the unit symbols do not contain multipliers and therefore represent the base derived unit to which a multiplier can be applied as a whole.

Every unit symbol is treated as an unparseable text as if it were a single-letter symbol. The meaning of each unit symbol is defined by the accompanying descriptive text and not by the text contents of the unit symbol.

To allow the widest possible range of serializations without requiring special character handling, several substitutions are made which deviate from the format described in IEC 80000-1. The division symbol "/" is replaced by the letters "Per". Exponents are written in plain text after the unit as "m^3". The letters "deg" are used instead of the degree symbol. Any clarification of the meaning for a substitution is included in the description for the unit symbol.

Non-SI units are included in list of unit symbols to allow sources of data to be correctly labelled with their non-SI units (for example, a GPS sensor that is reporting numbers that represent feet instead of meters). This allows software to use the unit symbol information correctly convert and scale the raw data of those sources into SI-based units.

The integer values are used for harmonization with IEC 61850.

A

A2

A2h

A2s
APerA
APerm
Ah
As
Bq
Btu
C
CPerkg
CPerm2
CPerm3

F
FPerm
G
Gy
GyPers
H
HPerm
Hz
HzPerHz
HzPers
J

JPerK

JPerkg

JPerkgK

JPerm2

JPerm3

JPermol

JPermolK

JPers

K

KPers

M

Mx
N
NPerm
Nm
Oe
Pa
PaPers
Pas
Q
Qh
S

SPerm
Sv
T
V
V2
V2h
VA
VAh
VAr
VArh
VPerHz

VPerV

VPerVA

VPerVAr

VPerm

Vh

Vs

W

WPerA

WPerW

WPerm2

WPerm2sr

WPermK

WPers

WPersr

Wb

Wh

anglemin

anglesec

bar

cd

charPers

character

`cosPhi`
`count`
`d`
`dB`
`dBm`
`deg`
`degC`
`ft3`
`gPerg`
`gal`
`h`

ha
kat
katPerm3
kg
kgPerJ
kgPerm
kgPerm3
kgm
kgm2
kn
l

lPerh

lPerl

lPers

lm

lx

m

m2

m2Pers

m3

m3Compensated

m3Perh

m3Perkg
m3Pers
m3Uncompensated
mPerm3
mPers
mPers2
min
mmHg
mol
molPerkg
molPerm3

molPermol

none

ohm

ohmPerm

ohmm

onePerHz

onePerm

ppm

rad

radPers

radPers2

rev	
rotPers	
s	
sPers	
sr	
therm	
tonne	

VsPpccControlKind

DC

Types applicable to the control of real power and/or DC voltage by voltage source converter.

pPcc	Control is real power at point of common coupling. The target value is provided by ACDCConverter.targetPpcc.

pPccAndUdcDroop	Control is active power at point of common coupling and local DC voltage, with the droop. Target values are provided by ACDCConverter.targetPpcc, ACDCConverter.targetUdc and VsConverter.droop.
pPccAndUdcDroopPilot	Control is active power at point of common coupling and the pilot DC voltage, with the droop. The mode is used for Multi Terminal DC (MTDC) systems where multiple DC substations are connected to the DC transmission lines. The pilot voltage is then used to coordinate the control the DC voltage across the DC substations. Targets are provided by ACDCConverter.targetPpcc, ACDCConverter.targetUdc and VsConverter.droop.
pPccAndUdcDroopWithCompensation	Control is active power at point of common coupling and compensated DC voltage, with the droop. Compensation factor is the resistance, as an approximation of the DC voltage of a common (real or virtual) node in the DC network. Targets are provided by ACDCConverter.targetPpcc, ACDCConverter.targetUdc, VsConverter.droop and VsConverter.droopCompensation.
phasePcc	Control is phase at point of common coupling. Target is provided by VsConverter.targetPhasePcc.
udc	Control is DC voltage with target value provided by ACDCConverter.targetUdc.

VsQpccControlKind

DC

Kind of reactive power control at point of common coupling for a voltage source converter.

powerFactorPcc	Control is power factor at point of common coupling. Target is provided by VsConverter.targetPowerFactorPcc.
pulseWidthModulation	No explicit control. Pulse-modulation factor is directly set in magnitude (VsConverter.targetPWMfactor) and phase (VsConverter.targetPhasePcc).
reactivePcc	Control is reactive power at point of common coupling. Target is provided by VsConverter.targetQpcc.
voltagePcc	Control is voltage at point of common coupling. Target is provided by VsConverter.targetUpcc.

WindingConnection

Wires

Winding connection type.

A	

D
I
Y
Yn
Z
Zn

Compound Types

Datatypes

ActivePower Domain

Product of RMS value of the voltage and the RMS value of the in-phase component of the current.

XSD type: **float**

ActivePowerPerCurrentFlow Domain

Active power variation with current flow.

XSD type: **float**

AngleDegrees Domain

Measurement of angle in degrees.

XSD type: **float**

AngleRadians Domain

Phase angle in radians.

XSD type: **float**

ApparentPower

Domain

Product of the RMS value of the voltage and the RMS value of the current.

XSD type: **float**

Capacitance

Domain

Capacitive part of reactance (imaginary part of impedance), at rated frequency.

XSD type: **float**

Conductance

Domain

Factor by which voltage must be multiplied to give corresponding power lost from a circuit. Real part of admittance.

XSD type: **float**

CurrentFlow

Domain

Electrical current with sign convention: positive flow is out of the conducting equipment into the connectivity node. Can be both AC and DC.

XSD type: **`float`**

Frequency

Domain

Cycles per second.

XSD type: **`float`**

Inductance

Domain

Inductive part of reactance (imaginary part of impedance), at rated frequency.

XSD type: **`float`**

Length

Domain

Unit of length. It shall be a positive value or zero.

XSD type: **`float`**

PU Domain

Per Unit - a positive or negative value referred to a defined base. Values typically range from -10 to +10.

XSD type: **`float`**

PerCent Domain

Percentage on a defined base. For example, specify as 100 to indicate at the defined base.

XSD type: **`float`**

Reactance Domain

Reactance (imaginary part of impedance), at rated frequency.

XSD type: **`float`**

ReactivePower Domain

Product of RMS value of the voltage and the RMS value of the quadrature component of the current.

XSD type: **`float`**

RealEnergy Domain

Real electrical energy.

XSD type: **`float`**

Resistance Domain

Resistance (real part of impedance).

XSD type: **`float`**

RotationSpeed Domain

Number of revolutions per second.

XSD type: **`float`**

Seconds

Domain

Time, in seconds.

XSD type: **float**

Susceptance

Domain

Imaginary part of admittance.

XSD type: **float**

Voltage

Domain

Electrical voltage, can be both AC and DC.

XSD type: **float**

VoltagePerReactivePower

Domain

Voltage variation with reactive power.

XSD type: **`float`**

Primitive Types

[*Boolean*]() Domain

A type with the value space "true" and "false".

XSD type: **`boolean`**

[*DateTime*]() Domain

Date and time as "yyyy-mm-ddThh:mm:ss.sss", which conforms with ISO 8601. UTC time zone is specified as "yyyy-mm-ddThh:mm:ss.sssZ". A local timezone relative UTC is specified as "yyyy-mm-ddThh:mm:ss.sss-hh:mm". The second component (shown here as "ss.sss") could have any number of digits in its fractional part to allow any kind of precision beyond seconds.

XSD type: **`dateTime`**

[*Float*]() Domain

A floating point number. The range is unspecified and not limited.

XSD type: **float**

Integer

Domain

An integer number. The range is unspecified and not limited.

XSD type: **integer**

String

Domain

A string consisting of a sequence of characters. The character encoding is UTF-8. The string length is unspecified and unlimited.

XSD type: **string**